UDCT v7.14-TN01: Phase-Space Admissibility Audit (p = 1.5, r_a/a ≈ 0.4665)

Independent Technical Note | Retrospective exploratory audit | 11 November 2026

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GitHub: https://github.com/wspaik5/UDCT_v7.14_TN01_Phase_Space_Admissibility_Audit

Publication note. This DOI identifies the Zenodo deposit and becomes registered upon publication. The current publication status is shown on the Zenodo record. This technical note is not a new preregistration or a confirmatory observational result.

Scope and source preservation

This release retains the complete 32-page source-grounded research draft, including its original figures, provenance notes, numerical tables, archive index and scientific caveats, as an unchanged appendix. The pages after this front matter retain their original draft heading and status; they are intentionally not represented as a newly executed independent verification. The new front matter is an editorial contextualization, not a substitute for the archived research record.

Two mathematically distinct exploratory observations

1. Central asymptotic cancellation, p = 1.5. For the specified isolated spherical simple-QUMOND/Plummer model with tracer family rho_p(r) proportional to [1+(r/a)^p]^(-5/p), the central potential difference scales as r^(3/2). The tracer-density expansion therefore has exponent s = 2p/3. At p = 1.5, s = 1 and the identified leading singular distribution-function inversion coefficient cancels. This does not establish the sign of the full distribution function at equality: higher-order and regular terms, boundary conventions, and full-domain checks remain necessary.

2. Local OM tangency, r_a/a ≈ 0.466466787396. In the specified Osipkov–Merritt anisotropy branch at p = 1.5, the archived calculation reports a finite-window zero-contact condition near r_a/a = 0.46646678739601644. The associated turning-point label is r/a ≈ 0.47409755, a distinct dimensionless quantity. This is a model-dependent sampled numerical boundary, not a universal constant, not a global positivity proof, and not an observed galactic radius.

Physical admissibility and limits

A negative phase-space distribution function would rule out the specified equilibrium distribution-function ansatz at that energy; it would not by itself falsify MOND, QUMOND, GQUMOND, or establish dynamical instability. Three-dimensional gravitational Hessian diagnostics, spherical Jeans second moments, and spherical distribution-function inversion are distinct tests. Hydrus I supplies motivation and selected inputs, but the present work is not a likelihood fit to all individual stellar line-of-sight velocities.

Future directions

Hydrus I: Replace illustrative tracer assumptions with constrained stellar profiles; test anisotropy and external-field sensitivity; perform star-by-star velocity likelihood comparisons, reporting selection and membership uncertainty.

Other dwarf galaxies: Determine whether analogous asymptotic cancellations or admissibility boundaries persist, shift, or disappear when changing tracer cusps, baryonic sources, environmental fields, and equilibrium assumptions. The numerical values $p = 1.5$ and $r_a/a \approx 0.4665$ are not assumed to be transferable constants.

Bullet Cluster: Develop a separate three-dimensional, time-dependent, non-equilibrium framework with evolving gas and galaxy mass distributions and lensing comparisons. The isolated spherical equilibrium OM threshold cannot be directly imposed on a colliding cluster; first define an appropriate non-equilibrium consistency observable and independently validate its implementation.

Confirmatory pathway: Derive the next nonzero equality-case term; check the full accessible energy domain; perform resolution and quadrature convergence tests; freeze new independent predictions and decision rules before execution; preserve complete code, inputs, logs, and failure cases.

Reproducibility boundary

This package includes a source-grounded document and a source-audit table, not a newly rerun physics solver. The 1,642 dialogue screenshots have not been exhaustively reviewed. A new independent implementation and raw archive verification are still required before claiming complete computational reproduction.

UDCT v7.14 — Phase-Space Admissibility Audit (p = 1.5; r_a/a ≈ 0.4665)

The p = 1.5 central asymptotic cancellation and the r_a/a ≈ 0.466466787396 finite-window tangency: source-based reconstruction, numerical diagnostics, and prospective falsification program

Won Shik Paik | Independent research archive | 10 November 2026 | SOURCE-GROUNDED DRAFT, NOT A FINAL COMPLETE 1,642-IMAGE REVIEW

SOURCE STATUS: 46 archived follow-up artifacts are indexed below; these are filename-sorted source items, not a verified chronological record of 46 distinct experiments. Earlier Steps 01–50 are distinct. This report reuses original computed CSV figures and archival summaries; no independent physics solver rerun or observational fit is claimed.

Abstract

This paper reconstructs the exploratory UDCT v7.14 investigation from original source archives, archived numerical outputs, and prior audit notes. In a specified isolated spherical simple-QUMOND/Plummer configuration, the central tracer exponent p = 3/2 cancels a leading asymptotic coefficient. A separate Osipkov–Merritt (OM) distribution-function calculation identifies a candidate local tangency near anisotropy radius ratio q = r_a/a = 0.46646678739601644 over a finite energy window. We explain why these are distinct mathematical observations, how the numerical audits constrain their interpretation, and why finite-grid agreement does not prove global non-negativity, dynamical stability, observational agreement, or priority over prior art. Four plots derived in the earlier figure reconstruction from archived CSVs are retained with expanded scientific interpretation. A complete 46-item archive index reproduces source descriptions and relevant original numerical snippets, preserving provenance. We define prospective tests for numerical convergence, distribution-function admissibility, dwarf-galaxy likelihoods, and time-dependent cluster lensing, explicitly separating model-internal tests from observational falsification.

1. Introduction and research question

A kinematic second-moment or rotation-curve fit does not guarantee the existence of a non-negative equilibrium stellar distribution function. The UDCT v7.14 archive therefore combines several logically different questions: three-dimensional Hessian diagnostics, spherical Jeans moments, the OM inversion of an augmented tracer density, and the numerical sensitivity of inferred sign changes. These questions should not be collapsed into a single claim of gravitational viability. The central objective here is to reconstruct exactly what was tested and to identify the next falsifiable calculations.

The expressions “phase shift” and “critical boundary” are used carefully: the present calculations do not demonstrate a time-dependent phase transition. They identify candidate boundaries in asymptotic coefficients and in a model-dependent finite-window sign calculation. In particular, the q ≈ 0.4665 candidate must not be treated as a universal astrophysical constant.

1.1. Dialogue-image cross-check: preliminary direct observations

The dialogue archive contains 1,642 JPEG screenshots. A cross-sectional visual sample spanning the archive was examined directly, but the full 1,642-image conversation has NOT been individually read. The sampled screenshots reinforce several important audit distinctions: one discussion explicitly separates the central p=1.5 coefficient cancellation from the OM local tangency; another reports a p=1.501 negative-DF region; another describes 36 local recomputations; and a later status table says global energy-domain positivity and direct Hydrus I validation remain unconfirmed. A screenshot also reports a Gemini numerical acceleration implementation error, so the dialogue cannot be treated as independently verified numerical evidence. These observations are leads to be reconciled against code and logs, not confirmations of the quoted calculations.

The screenshot chronology cannot be reconstructed merely from the final screenshot export time: filenames contain both apparent original capture times and later export timestamps, and some screenshots duplicate or summarize earlier conversations. The source-image audit must distinguish proposal, code execution, output, later commentary, and

contradictory interpretation before treating any statement as a scientific result.

2. Source corpus, provenance and chronology

The integrated package includes two original Steps 01–25 and 26–50 ZIP archives, six early exploratory audits, a 46-item follow-up collection, reproducibility bundles, and a prospective protocol. The Phase I source audit reports CRC and SHA-256 provenance checks, including byte-identical embedded originals, but expressly does not claim a fresh solver execution or independent physical validation. Some records are duplicated or reissued; item numbers in the appendix follow the archive ledger rather than execution time. A 1,642-image dialogue archive was provided but has not been fully transcribed or individually cross-indexed here; no claim of exhaustive conversation reconstruction is made.

The prospective protocol in the bundle is not evidence of prospective confirmation. Its 20-point grid is explicitly not verified as a never-before-explored holdout. Future confirmatory work requires a frozen protocol and a genuinely independent implementation before running new tests.

3. Fixed mathematical model and conventions

The archived gravitational source is a spherical Plummer profile, with g_N(r) = G M r/(r²+a²)^(3/2). The isolated spherical simple-QUMOND algebraic field is g(r) = g_N(r)/2 + sqrt[g_N(r)²/4 + a_0 g_N(r)]. The tracer density is rho_p(r) = rho_0 [1+(r/a)^p]^(-5/p). The Osipkov–Merritt augmented density is rho_Q(r) = [1+(r/r_a)²]rho_p(r), with beta(r) = r²/(r²+r_a²). The source protocol fixes G = 4.30091×10^-3 pc (km/s)²/Msun, M = 32730 Msun, a = 53.3 pc, and a_0 = 1.2×10^-10 m/s². These are inputs, not fitted observational predictions.

Potential conventions are crucial. The outward-increasing difference Phi(r)-Phi(0) is used, because the outer isolated-MOND potential grows logarithmically and cannot simply be assigned zero at infinity. The archived energy label x = r/a denotes a turning-point label in the spherical inversion, not a measured stellar position. Any comparison of two inversion formulas must use identical potential zero, boundary terms, density normalization, and energy definitions.

4. Central asymptotic structure at p = 1.5

At small x = r/a, the archived source note expands g(x) = h x^(1/2) + (n0/2)x + …, with n0 = GM/a² and h = sqrt(a_0 n0). Integrating yields Phi(x)-Phi(0) = B x^(3/2) + E x² + …, B = 2ah/3, E = an0/4. For p = 3/2 the augmented tracer density has rho_Q = 1 -(10/3)x^(3/2) + q^(-2)x² + … . The leading asymptotic coefficient in the inversion branch vanishes at p = 1.5. The cancellation is analytic within the specified assumptions, but at equality the subleading x² contribution becomes essential; the sign of the full DF does not follow from the vanishing coefficient alone.

A source central-subtraction audit reports B = 15.220461348913249, E = 0.6602663428705441, the augmented-density x² coefficient 4.740373345250881, and the kernel coefficient A = 0.00909440720938013 for the fixed model and q candidate. Three finite-segment direct-versus-subtracted integrals differ by approximately 6.47×10^-15, 1.96×10^-15, and 1.57×10^-16 in archived outputs. Because the two paths share an algebraic kernel, these differences demonstrate internal numerical consistency on the tested segments, not independent physical replication or certified global error bounds.

5. Candidate OM tangency at q ≈ 0.4665

The second numerical feature is the reported local tangency near q = 0.46646678739601644 for p = 1.5. Its interpretation requires identifying the tested energy window, the minimum of the reconstructed OM DF, and the numerical tolerance. A minimum near zero in a sampled window is not a proof that f(Q) ≥ 0 at every physically accessible energy. Conversely, a certified negative value at any admissible energy would exclude the corresponding assumed equilibrium DF, not automatically all MOND theories.

The OM anisotropy radius q is not the same as a Hessian sign, a stability eigenvalue, or a physical phase-transition coordinate. The earlier 3D Hessian and Jeans studies motivated this branch but do not establish the OM inversion result. No claim is made here that these computations implement the full higher-derivative GQUMOND action.

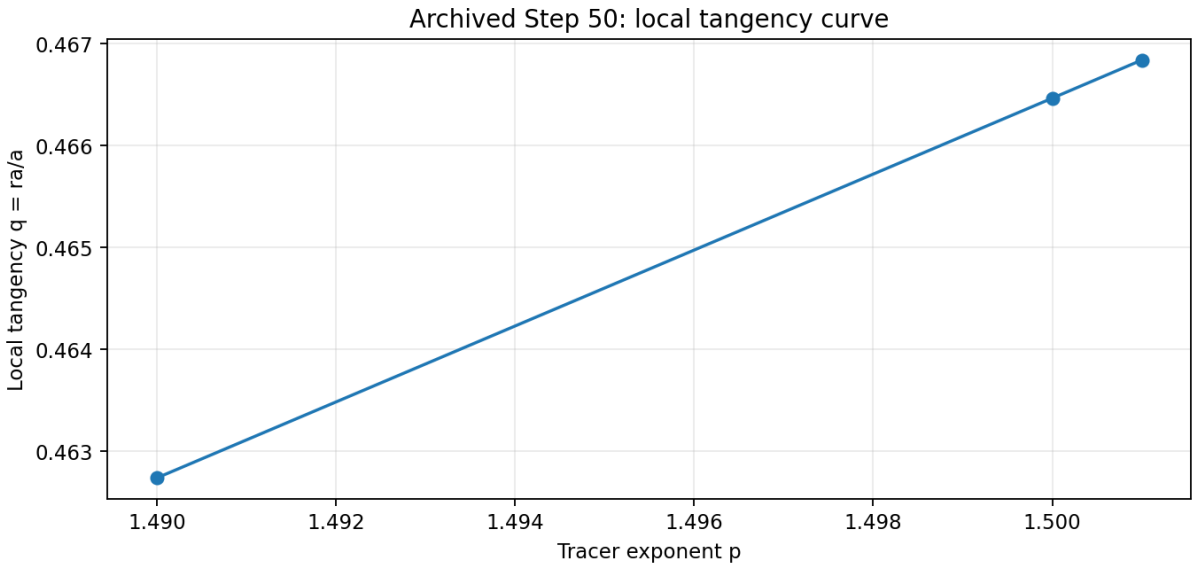
6. Numerical methodology, controls and uncertainty

The archives describe finite-energy scans, adaptive quadrature, Gauss orders 64/128/256, squared-log endpoint transforms, central regularity gates, outer-tail tests, independent-code attempts, Newtonian OM controls, and alternative integration-by-parts expressions. Each tests a different possible failure mode. Successive quadrature differences are empirical convergence diagnostics rather than rigorous error enclosures; methods that share a potential or kernel are not physically independent. Near a zero crossing, the sign decision must use a bound on the total error that is demonstrably smaller than the absolute inferred DF value.

The central r/a < 10^-9 exclusion reported in an earlier regularity gate and finite outer cutoffs are explicit limitations. The Q = 0 endpoint, logarithmic outer potential, and subleading central terms must be handled before any global positivity assertion. A three-valued decision rule—positive, negative, or unresolved—should be used rather than assigning a sign to values consistent with numerical zero.

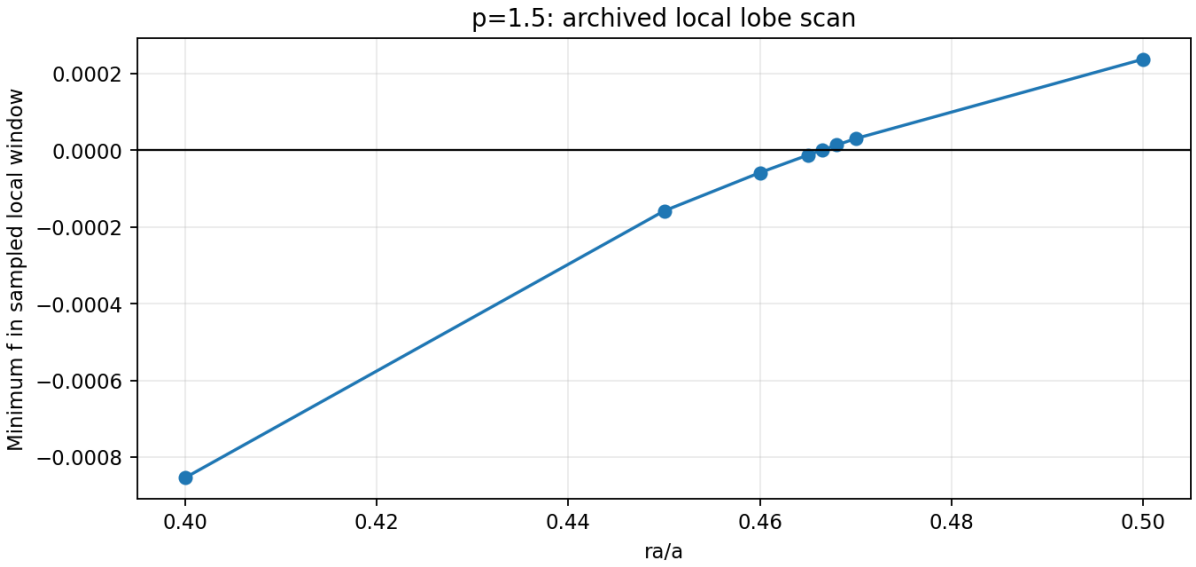
7. Archived figures and interpretation

Figure 1. Candidate tangency landscape versus tracer exponent



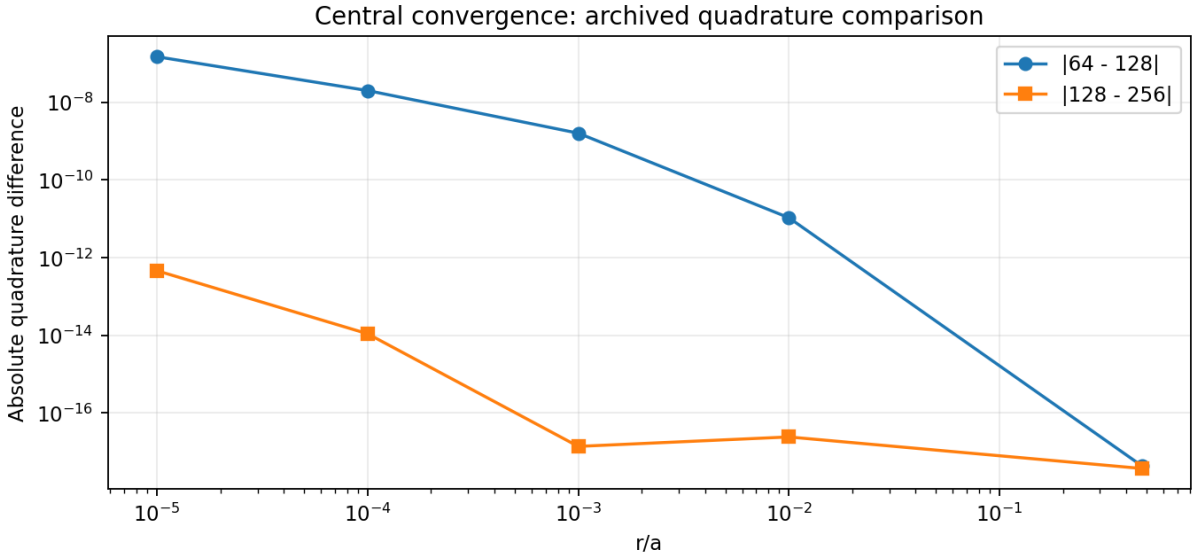
This archived CSV-derived plot compares candidate anisotropy thresholds as the tracer exponent varies. A plotted root or tangency is only as robust as its finite-energy minimization and error control. The scientific test is whether the same feature persists under a denser p–q–energy grid and an independently implemented inversion. Do not interpret the plotted curve as a universal phase boundary.

Figure 2. Finite-window minimum and the q ≈ 0.4665 crossing



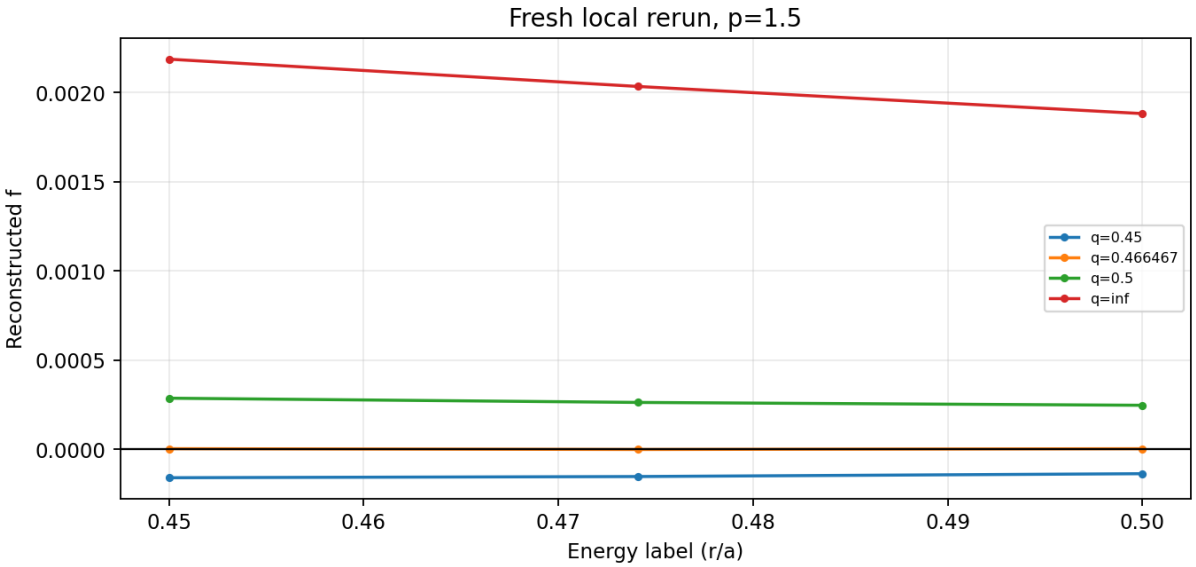
The minimum of the reconstructed distribution function is evaluated as the OM anisotropy scale changes. Values near zero are most sensitive to quadrature and energy-grid resolution. An apparent sign change must be checked with certified uncertainty, especially near the local tangency, and must not be extrapolated beyond the tested energy interval.

Figure 3. Quadrature order sensitivity



Comparing 64-, 128- and 256-node results reveals whether the numerical estimate stabilizes under a particular integration strategy. Agreement of successive orders is useful evidence against gross discretization error, but not a rigorous error bound, and not a cross-check of the physical field equation itself.

Figure 4. Local p = 1.5 rerun across anisotropy settings



The local rerun illustrates how a candidate sign or near-zero region changes across nearby q values. A finite set of points can miss narrow negative intervals. The follow-up must use adaptive energy minimization, root isolation and explicit endpoint treatment.

8. Results synthesis and interpretation

The strongest supported result is the model-conditional central asymptotic cancellation at p = 1.5 and a separately localized numerical OM tangency candidate near q = 0.4665. Their combination provides a useful stress-test design: one feature controls a central expansion, while the other tests numerical phase-space admissibility in a finite energy domain. The archival record does not establish a new universal MOND law, observational validation, a globally positive DF, or a claim of scientific priority. The scientific value lies in narrowing testable assumptions and recording failures as well as successes.

9. Prospective verification and falsification plan

9.1 Controlled numerical replication

Freeze the field equation, potential reference, Plummer parameters, tracer exponent, anisotropy definition, energy variable, and boundary terms. Publish source hashes and an untouched validation grid. Implement the inversion in two independently written codebases and two analytic representations, explicitly retaining the Q=0 boundary term. Compare dense adaptive minimization of f(Q) with high-precision interval or validated error calculations; classify positive, negative or unresolved. Report the worst-case energy location, error envelope, and any failed convergence gates. The existing proposed 20-point grid $p \in \{1.487,1.513\}$, $q \in \{0.44,0.48\}$, $x \in \{0.30,0.40,0.50,0.60,0.70\}$ is a documented future grid but not a verified blind holdout.

9.2 Dwarf galaxies: Hydrus I and replication targets

Replace illustrative model inputs with a published selection function, distance, light profile, membership model, and individual line-of-sight velocity likelihood. Propagate uncertainties in binaries, contaminants, external field, anisotropy, and possible tides. Solve the gravitational field for each predeclared model; construct physically admissible DFs, project to observable velocity distributions, and compare against matched Newtonian dark-matter and MOND baselines using out-of-sample predictive checks. A phase-space failure excludes the specified equilibrium assumptions; a velocity-likelihood failure challenges the observational model; neither alone establishes a universal theory-level falsification. Hydrus I-informed inputs in the existing archive are not equivalent to a direct fit of all individual stellar velocities.

9.3 Bullet Cluster: non-equilibrium and lensing

Do not insert the spherical q = 0.4665 value into a colliding cluster. Build 3D gas and galaxy mass maps with observational uncertainty, solve Newtonian and specified modified-gravity fields, and evolve matched collision initial conditions including merger geometry and relative velocities. Use a justified relativistic lensing prescription to predict convergence and shear, since a nonrelativistic QUMOND potential alone does not determine lensing. Compare centroid offsets, mass-map morphology, gas shocks, galaxy velocities, and lensing likelihoods. Equilibrium OM inversion is not directly valid for a time-dependent merger: phase-space consistency must instead be assessed through time-dependent collisionless evolution and positivity of the evolved DF. Any claim of MOND-wide exclusion requires ruling out the relevant full model class, including explicitly stated extra degrees of freedom.

9.4 Interpretation matrix

If both boundaries reproduce with independent error control, they become verified constraints of the specified model. If the local tangency moves with cutoff, numerical method, or tracer assumptions, it remains model-dependent. If a globally admissible DF and good dwarf-galaxy likelihood coexist, the model survives that test only. If the cluster fails after all predeclared alternatives and uncertainty budgets are evaluated, the tested framework is challenged at cluster scales. If a DF becomes robustly negative under frozen assumptions, that equilibrium construction is rejected. All outcomes—including inconclusive cases—must be published.

10. Limitations, novelty and conclusion

Non-negative DFs, OM anisotropy, and inversion positivity are established subjects in stellar dynamics. Novelty cannot be claimed solely from using these concepts or reporting many decimal places. A defensible contribution would be a new, independently reproduced boundary in a clearly specified QUMOND model, its analytic origin, and a source-complete record of failures and controls. The present study is an archival synthesis with candidate results and a testable program, not proof of a universal physical phase shift or a decisive test of dark matter versus MOND.

The central result is a methodology for turning apparently successful kinematic models into stronger, falsifiable phase-space tests. The next research milestone is an independently coded, error-controlled inversion over a predeclared domain; only then should the program proceed to likelihood-based dwarf observations and non-equilibrium cluster dynamics.

References and source documentation

Bekenstein & Milgrom (1984), Does the missing mass problem signal the breakdown of Newtonian gravity?, Astrophysical Journal 286, 7–14.

Milgrom (2010), Quasi-linear formulation of MOND, Monthly Notices of the Royal Astronomical Society 403, 886–895.

Osipkov (1979), Spherical systems of gravitating bodies with an ellipsoidal velocity distribution, Soviet Astronomy Letters 5, 42.

Merritt (1985), Spherical stellar systems with spheroidal velocity distributions, Astronomical Journal 90, 1027.

Binney & Tremaine (2008), Galactic Dynamics, 2nd ed., Princeton University Press.

UDCT v7.14 archived Steps 01–50, six exploratory audits, 46-item follow-up ledger, and integrated Phase I source audit (user-provided archives, 2026).

UDCT v7.14 Prospective Numerical Phase-Space Consistency Preregistration (archived protocol, 10 November 2026; status must be checked before citing as published).

9.5 Future directions: transferring the p = 1.5 and r_a/a ~ 0.4665 diagnostics

The two exploratory markers are not interchangeable physical constants. In the specified spherical model, p = 1.5 marks cancellation of a leading central asymptotic coefficient; it does not determine the sign of the complete distribution function at equality. The value r_a/a ~ 0.4665 is a local finite-window Osipkov-Merritt (OM) tangency candidate for a particular source, tracer, anisotropy prescription and numerical domain. Neither is currently established as a universal galaxy or cluster threshold.

Hydrus I: observation-linked tests

Reconstruct the stellar tracer density, gravitational potential and external-field assumptions from published Hydrus I data, including membership and selection effects. Test whether the central-expansion cancellation has an analogue near p = 1.5 after fitting or marginalizing over the tracer profile; recompute any OM tangency rather than fixing r_a/a = 0.4665. First perform model-internal Jeans and full-energy distribution-function admissibility checks. Only in a separate observational stage, compare predicted line-of-sight velocity distributions with individual-star measurements and uncertainty-aware likelihoods.

Other dwarf galaxies: replication and falsification

Select independent dwarf-galaxy targets spanning stellar concentration, anisotropy, tidal environment and external-field strength. Apply the same frozen definitions, energy-domain and convergence checks to each galaxy, but allow its observed structural scales and inferred tracer exponent to vary. Record whether a p = 1.5-like cancellation, an OM tangency, or neither occurs. Compare with matched Newtonian dark-matter and MOND/QUMOND baselines and report null results, shifts in the boundaries, and unresolved cases without interpreting model-specific failures as universal falsification.

Bullet Cluster: non-equilibrium generalization, not direct transplantation

The Bullet Cluster is a time-dependent, non-spherical merger; the spherical equilibrium OM parameter r_a/a and its numerical value ~0.4665 cannot simply be transferred. Instead, test whether mathematically analogous regularity, phase-space nonnegativity or field-derivative constraints survive in explicitly specified three-dimensional merger models. Evolve gas and collisionless components from matched initial conditions; predict galaxy and gas dynamics and, with a justified relativistic prescription, lensing observables. Compare against measured mass-map morphology and lensing centroid offsets. Any link to p = 1.5 must be derived anew for a clearly defined cluster-scale density family, not presumed from the dwarf model.

Predeclared decision rules and research status

For all targets, preregister the field equation, parameter ranges, observable definitions, numerical tolerances, uncertainty budget and pass/fail/inconclusive rules before confirmatory runs. Preserve the current p = 1.5 and r_a/a ~ 0.4665 observations as exploratory motivation, not independent prospective predictions. No Hydrus I star-by-star likelihood fit, cross-galaxy validation, or Bullet Cluster merger/lensing simulation is claimed as completed by this technical note.

Appendix A. Complete 46-item source ledger with original technical excerpts

Entries below reproduce technical-note and README snippets from the prior source reconstruction. This is an artifact ledger, not 46 confirmed unique simulations or a verified chronology. Text labeled ORIGINAL is preserved from archive extracts; it is not independently verified physics. Long outputs are clipped per item for legibility; the companion Markdown retains fuller extracts.

Archive item 1: UDCT_v714_Central_Asymptotic_Subtraction_REISSUED_20261110_v2_20261110_125051_841.zip

- SHA-256: `f08d9bd738b1c2c3cd28fa5c99b2438e9751123b50e6b218908b64c7f0c5a803`; bytes: 5482; nested CRC: pass

- Contained files: `README.md`, `SHA256SUMS.txt`, `TECHNICAL_NOTE.md`, `audit.py`, `kernel_asymptotic.csv`, `requirements.txt`, `subtraction_segments.csv`, `summary.json`

Source excerpt — README.md (excerpt):

Reproducibility — Central Asymptotic Subtraction

Run `pip install -r requirements.txt` and `python audit.py`. It regenerates `kernel_asymptotic.csv`, `subtraction_segments.csv`, and `summary.json`. This is an exploratory follow-up to the Central Convergence Audit; it is not preregistered

Source excerpt — TECHNICAL_NOTE.md (excerpt):

UDCT v7.14 — Central Asymptotic Subtraction (Exploratory)

Model and derivation

For x=r/a and q=r_a/a, the Plummer-source algebraic MOND field has g(x)=h x^(1/2)+(n0/2)x+..., where n0=GM/a² and h=sqrt(a0 n0). Hence Phi(x)-Phi(0)=B x^(3/2)+E x²+..., B=2ah/3, E=an0/4. For p=3/2, rho_Q=(1+x^(3/2))^(-10/3)(1+x²/q²)=1-(10/3

Numerical check

The derived A/x is compared to the exact differential kernel in kernel_asymptotic.csv. A finite central segment is integrated directly and independently as the sum of a numerical integral of the analytic leading asymptotic expression and a

Scope

This audit isolates the leading central term for positive-energy (nonzero lower-radius) segments. It does not establish a closed-form finite-segment integral, a rigorous numerical error enclosure, the behavior of f(Q) at Q=0, or global posi

Source excerpt — summary.json (excerpt):

{"analytic_B": 15.220461348913249, "analytic_E": 0.6602663428705441, "analytic_augmented_density_x2_coefficient": 4.740373345250881, "analytic_kernel_coefficient_A": 0.00909440720938013, "model": "isolated spherical algebraic MOND field, Pl

Archive item 2: UDCT_v714_Central_Convergence_Audit_20261110_20261110_125043_851.zip

- SHA-256: `51386f2b280587df8929f4a2c93ddce646b99ef1bef85acbc67c369351a39056`; bytes: 4409; nested CRC: pass

- Contained files: `audit.py`, `convergence.csv`, `summary.json`, `TECHNICAL_NOTE.md`, `README.md`, `requirements.txt`, `SHA256SUMS.txt`

Source excerpt — TECHNICAL_NOTE.md (excerpt):

UDCT v7.14 — Central Quadrature Convergence Audit (Exploratory)

This audit holds the original Plummer gravitational source, p=1.5 tracer, and OM ratio fixed. It computes the inversion integral using direct adaptive potential integration at each Gauss node, a squared-log-radius endpoint transform, and qu

Scope: convergence screening at four central radii and the candidate tangency. Differences between successive orders are empirical and not mathematically certified error bounds. No asymptotic subtraction was performed, and this audit mu

Results: see convergence.csv and summary.json.

Source excerpt — README.md (excerpt):

Reproduction

Run `python audit.py` after `pip install -r requirements.txt`. The script regenerates `convergence.csv` and `summary.json`. The previous 21-point scan is not modified. This is exploratory, not a preregistered confirmatory test.

Source excerpt — summary.json (excerpt):

{"model": "isolated spherical MOND interpolation, Plummer source, p=1.5 tracer, OM ra/a=0.46646678739601644", "radii": [1e-05, 0.0001, 0.001, 0.01, 0.474097553], "orders": [64, 128, 256], "outer_over_a": 1000000.0, "status": "exploratory nu

Archive item 3: UDCT_v714_Central_Regularity_Gate_REISSUED_20261110_v2_20261110_125028_970.zip

- SHA-256: `f927d13d974186d5943b88f652e19a157d4eff167d6c7a6ff69041aa84180379`; bytes: 8013; nested CRC: pass

- Contained files: `README.md`, `SHA256SUMS.txt`, `TECHNICAL_NOTE.md`, `audit.py`, `central_samples.csv`, `central_segments.csv`, `model.py`, `requirements.txt`, `run.log`, `summary.json`

Source excerpt — README.md (excerpt):

Central regularity gate — reproducibility

This is an exploratory numerical audit, **not** formal preregistration or global positivity proof.

Run:

~~~

python -m pip install -r requirements.txt

python audit.py

~~~

Outputs: `central_samples.csv`, `central_segments.csv`, `summary.json`. Source model is copied into `model.py` for portability. The truncated centre integrals deliberately omit `r/a < 1e-9` and must not be mistaken for finite endpoint value

Source excerpt — TECHNICAL_NOTE.md (excerpt):

UDCT v7.14 — Central regularity and energy-domain exploratory gate

Status: exploratory follow-up, not a preregistration result or global phase-space proof.

Model

Spherical Plummer gravitational source, isolated MOND interpolation as implemented in `model.py`, tracer density proportional to `[1+(r/a)^1.5]^(-5/1.5)`, Osipkov–Merritt augmented density `(1+r²/r_a²)rho(r)`, candidate `r_a/a=0.46646678739

Local limits

The gravitational field behaves as `g~r^(1/2)`, and `Phi(r)-Phi(0)~r^(3/2)`. Numerically, the OM kernel `d²rho_Q/dPhi²` behaves approximately as `r^(-1)` for the implemented model at p=1.5. The *log-radius* integrand of the formal zero-ener

The earlier local `p=1.5` cancellation does not imply that the *entire* inversion kernel is regular or vanishes at the centre: subleading terms matter.

Energy domain

The isolated MOND potential has finite central value and logarithmically grows to +infinity at large radii. A natural energy coordinate is `Q=Phi(r0)-Phi(0)` for `r0>=0`; the formal domain is `Q in [0,infinity)`. For any `Q>0`, `r0>0` and t

Numerical methods and limits

The script samples 26 central radii `1e-7 <= r/a <= 1e-2`, estimates log slopes on the smallest 11 points, and integrates a formal `Q=0` logarithmic-radius integrand from `1e-9 a` to four outer cutoffs. Adaptive quadrature's reported errors

Source excerpt — summary.json (excerpt):

{"q": 0.46646678739601644, "tracer_p": 1.5, "small_radius_log_slopes": {"gravity": 0.5000337289602065, "kernel_d2rhoQ_dPhi2": -1.0000880359573676, "phi_minus_phi0": 1.500025111842174, "zero_energy_log_integrand": -0.2500668629182483}, "segm

Archive item 4: UDCT_v714_Central_Subtraction_Convergence_REISSUED_20261110_v2_20261110_125057_940.zip

- SHA-256: `4d96a1e91adf0a7d64353596500d6ce0abdc7d4b333a89d7a8ca8de6beb686b1`; bytes: 6424; nested CRC: pass

- Contained files: `audit.py`, `convergence.csv`, `summary.json`, `TECHNICAL_NOTE.md`, `README.md`, `requirements.txt`, `SHA256SUMS.txt`

Source excerpt — TECHNICAL_NOTE.md (excerpt):

UDCT v7.14 — Central Asymptotic Subtraction: Quadrature-Order Convergence Audit

Status: exploratory numerical audit, 2026-11-10. Not a preregistered confirmatory result.

Objective

Compare fixed-order Gauss–Legendre quadrature on three finite central radial segments for the same OM inversion kernel, (a) directly and (b) after subtraction of the analytically derived leading central term. The comparison reference is ord

Model

Spherical Plummer gravitational source with algebraic MOND field; tracer density proportional to `(1+x^(3/2))^(-10/3)`; OM augmented density multiplied by `(1+x^2/q^2)`, with `x=r/a` and `q=0.46646678739601644`. Parameters and units match t

The central field has `g ~ h sqrt(x)` and `Phi-Phi(0) ~ B x^(3/2)`, where `h=sqrt(a0 GM/a^2)` and `B=2ah/3`. For `p=3/2`, the augmented density yields `d²rho_Q/dPhi² ~ A/x`, with `A=4[q^-2+(10/3)E/B]/(9B²)` and `E=a(GM/a²)/4`.

Procedure

For each finite interval `[x0,xc]`, transform `x=x0+(xc-x0)t²`, `0<=t<=1`. Calculate the direct integral using fixed-order Gauss–Legendre nodes, and calculate the residual after subtracting the leading integrand at the same nodes. Compute t

The potential difference `Phi(x)-Phi(x0)` is computed directly as an interval-local adaptive quadrature, avoiding subtraction of large separately evaluated potentials.

Results

| Finite segment x0..xc | Direct abs(16−256) | Subtracted abs(16−256) |

|---|---:|---:|

| 1e-6..1e-3 | 4.4205340451e-5 | 7.5913356157e-10 |

Source excerpt — README.md (excerpt):

UDCT v7.14 — Central Subtraction Convergence (Exploratory)

Run from a clean Python environment:

~~~bash

python -m pip install -r requirements.txt

python audit.py

~~~

Outputs: `convergence.csv` (15 rows) and `summary.json`. This audit compares direct Gauss–Legendre integration with leading-term-subtracted integration at orders 16, 32, 64, 128 and 256, on three finite central radial intervals.

The package does not demonstrate global distribution-function positivity or a universal critical OM radius. It does not replace the original Step 01–50 archives or earlier audit ZIPs.

Source excerpt — summary.json (excerpt):

{"model": "Plummer-source algebraic MOND, p=1.5 tracer, OM q=0.46646678739601644", "definition": "finite central segment only; leading reference via scipy quad, direct and residual via fixed-order Gauss-Legendre", "segments": [{"x0": 1e-06,

Archive item 5: UDCT_v714_Convergence_Audit_FRESH_20261031_v2_20261110_124829_235.zip

- SHA-256: `bdf559b5bdfea7a7e6bfb1ce2023825d78c55a248b2b5131d4012a058bf6ed5f`; bytes: 5248; nested CRC: pass

- Contained files: `README.md`, `convergence.csv`, `requirements.txt`, `run.log`, `run_convergence.py`, `step47_reference.py`, `summary.json`

Source excerpt — README.md (excerpt):

UDCT v7.14 — cutoff and quadrature convergence audit

Exploratory follow-up; does not modify the frozen v7.14 paper.

Run

`python -m pip install numpy pandas scipy`

`python run_convergence.py`

This script uses the archived Step 47 gravity/rho2phi kernel in `step47_reference.py` and computes 7 energy-coordinate radii × 3 finite outer cutoffs × 3 Gauss-Legendre orders (63 evaluations). Outputs `convergence.csv` and `summary.json`.

The coordinate `r_over_a` parameterizes energy via the potential at that radius. The `outer_over_a` value is the finite cutoff in the Step 47 numerical integral. The results are finite-sample convergence evidence, NOT proof that f(Q) is non

Next: adaptive minimization over continuous energy-coordinate subintervals, rigorous tail bounds, independent differentiation/inversion and forward reconstruction.

Source excerpt — summary.json (excerpt):

{"p": 1.5, "ra_over_a": 0.466466787396, "samples": 21, "min_f384": 5.572739167674104e-15, "negative_f384_below_minus_1e_10": 0, "max_abs_384_minus_192": 8.967829256967264e-11, "scope": "finite sampled radii, cutoff and quadrature convergenc

Archive item 6: UDCT_v714_Extended_Energy_Scan_REISSUED_20261030_20261110_124801_469.zip

- SHA-256: `e836c2e80ac142473a0920c8bf235c66bd5f110283c71e487942b6896dd9c672`; bytes: 5358; nested CRC: pass

- Contained files: `README_EXTENDED.md`, `run_extended.py`, `step47_reference.py`, `extended_scan.csv`, `extended_results.json`, `requirements.txt`

Source excerpt — extended_results.json (excerpt):

{"64": {"samples": 48, "minimum": 6.714803448654253e-15, "min_energy_label": 0.474097647, "negative_below_1e-10": 0, "negative_strict": 0}, "128": {"samples": 48, "minimum": 9.621032832818399e-15, "min_energy_label": 0.474097647, "negative_

Archive item 7: UDCT_v714_Extended_OM_Scan_20261109_20261110_124933_562.zip

- SHA-256: `6f258d9e9277f0114b350a8ee7fbe731ff33813b028f50ec0e31335309188b58`; bytes: 6772; nested CRC: pass

- Contained files: `README.md`, `extended_scan.csv`, `extended_scan.py`, `independent_check.py`, `requirements.txt`, `run.log`, `summary.json`

Source excerpt — README.md (excerpt):

UDCT v7.14 — Extended OM energy-label scan

Classification: exploratory, post-hoc numerical audit; not a preregistered confirmatory result.

Purpose

Extend the Step 50 local tangency scan from r_energy/a in [0.25,0.75] to [0.03,30] without changing the p=1.5 tracer or spherical Plummer source. Test whether the previously reported local OM tangency candidate ra/a=0.46646678739601644 is c

Execution

Python 3.10+ with numpy and scipy installed:

python extended_scan.py

Produces extended_scan.csv and summary.json. The archived run.log records a successful run. `independent_check.py` is a copy of the independently written adaptive-quadrature implementation from the previous tangency audit, retained for prov

Results

28 sampled energy labels across 0.03 to 30 a. At the original candidate ra/a, no sampled value of the computed inversion kernel was negative. The largest *sampled* critical ratio is approximately 0.466466787394 at r_energy/a about 0.4740975

Interpretation and critical caveats

- The coordinate `r_energy_over_a` labels the lower integration radius, not a globally normalized physical binding energy. In particular, isolated deep-MOND potentials have nontrivial asymptotic boundary conditions.

- The computation uses a finite upper integration radius (1e6 a). No independent verification of the correct physical boundary term or outer truncation has been performed.

- 28 samples cannot establish positivity over a continuum. The existing local maximum could be exceeded in unsampled narrow windows, near the center, or beyond r/a=30.

- The previous p=1.5 leading-order asymptotic cancellation is distinct from this OM local tangency.

Source excerpt — summary.json (excerpt):

{"status": "exploratory, post hoc; NOT preregistered", "source": "independent_check.py from earlier independent tangency audit", "p": 1.5, "outer_over_a": 1000000.0, "candidate_ra_over_a": 0.46646678739601644, "r_energy_label_range": [0.03,

Archive item 8: UDCT_v714_FD_Step_Convergence_REISSUED_20261110_v2_20261110_125111_151.zip

- SHA-256: `f5a85b21d8eced2234897166784aa78f31b385b929fb1d6a464ce34a6f6f91ef`; bytes: 7666; nested CRC: pass

- Contained files: `README.md`, `SHA256SUMS.txt`, `TECHNICAL_NOTE.md`, `audit.py`, `fd_sweep.csv`, `model.py`, `requirements.txt`, `run.log`, `summary.json`

Source excerpt — README.md (excerpt):

UDCT v7.14 — FD Step Convergence (Exploratory)

This package runs 63 finite-difference comparisons across 9 physical evaluation points. It compares analytic and finite-difference derivatives within the **same** OM inversion; this is **not** a fully independent OM inversion and does not e

Run: `python -m pip install -r requirements.txt` then `python audit.py`.

Contents: audit.py, model.py, fd_sweep.csv, summary.json, TECHNICAL_NOTE.md, requirements.txt, run.log, SHA256SUMS.txt.

Source excerpt — TECHNICAL_NOTE.md (excerpt):

UDCT v7.14 — Finite-Difference Step-Size Convergence Audit

Status: exploratory numerical audit, not preregistration and not a proof of phase-space positivity.

Scope and method

The existing Plummer gravitational-source / p=1.5 tracer / Osipkov–Merritt model and the existing direct-potential OM integral are retained. The analytic second derivative of augmented density with respect to potential is compared with a se

Three radial energy labels r/a = 0.3, 0.474097553, 0.7 and three anisotropy ratios ra/a = 0.4663667874, 0.46646678739601644, 0.4665667874 are crossed. Seven finite-difference increments h = 1e-2, 3e-3, 1e-3, 3e-4, 1e-4, 3e-5, 1e-5 are used.

Main finding

At r/a=0.474097553, ra/a=0.46646678739601644, analytic value = 1.7358936542081927e-14; finite difference h=1e-4 gives 5.19114206023041e-12, h=1e-5 gives 6.046896028435014e-14. Absolute discrepancy drops from 5.173783123688329e-12 to 4.31100

Interpretation and limitations

The signs either side of the candidate remain opposite in this finite sampling. The exact tangency remains indeterminate because the candidate value is extremely close to zero. This audit neither proves a global threshold nor bounds the con

Reproduction

Install requirements.txt and run `python audit.py` from this directory. Output: fd_sweep.csv, summary.json, run.log (when stdout is redirected). The copied model.py is the source used by this audit. Original

Source excerpt — summary.json (excerpt):

{"n_conditions": 9, "n_evaluations": 63, "model": "Plummer gravitational source; p=1.5 tracer; OM anisotropy; original MOND interpolation", "scope": "Finite-difference step convergence with shared potential and inversion integral; not indep

Archive item 9: UDCT_v714_Finite_Minima_REISSUED_20261031_v2_20261110_124836_178.zip

- SHA-256: `6b2759813aba6f35a9a7250870eef76650e24d0830bc5d53c86bc8c92e1bb42a`; bytes: 6095; nested CRC: pass

- Contained files: `run.py`, `step47_reference.py`, `global_grid.csv`, `minima.json`, `run.log`, `README.md`, `requirements.txt`

Source excerpt — README.md (excerpt):

UDCT v7.14 finite-domain minima audit

Run `pip install numpy scipy pandas` and `python run.py`.

Uses unmodified Step 47 physical kernel copied from the earlier source archive. This is NOT an independent physical inversion and NOT a proof of global positivity. The optimization brackets are finite; bounded minimization can miss multiple

Source excerpt — minima.json (excerpt):

[{"p": 1.499, "bracket": [0.25, 0.75], "rr": 0.47503173613918515, "f96": 3.253626215927443e-06, "f192": 3.2536262169968704e-06, "f384": 3.2536262261788922e-06, "success": true}, {"p": 1.499, "bracket": [0.001, 0.25], "rr": 0.249999990239410

Archive item 10: UDCT_v714_Finite_Outer_Tail_Audit_REISSUED_20261110_v2_20261110_125018_596.zip

- SHA-256: `31ef6026d7efcb469d377d11aff3ddbb74213e1915cc38c82a24a91cdb43164e`; bytes: 6961; nested CRC: pass

- Contained files: `README.md`, `SHA256SUMS.txt`, `TECHNICAL_NOTE.md`, `audit.py`, `model.py`, `requirements.txt`, `summary.json`, `tail_results.csv`

Source excerpt — README.md (excerpt):

UDCT v7.14 — Finite Outer-Tail Audit

Exploratory continuation of the OM outer-boundary asymptotic audit. Contains the archived direct-tangency reference model, executable tail audit, 36-row CSV, JSON summary, and technical note. Does not constitute formal preregistration.

Run: `python -m pip install -r requirements.txt` then `python audit.py`.

Python 3; NumPy and SciPy required. Units and normalization are inherited from `model.py`. No claims of certified error bounds or global DF positivity.

Source excerpt — TECHNICAL_NOTE.md (excerpt):

UDCT v7.14 — Finite Outer-Tail Audit (Exploratory)

Scope

This calculation uses the archived direct-tangency `model.py` (isolated spherical Plummer source, p=1.5 tracer, OM augmentation). It evaluates the contribution to the absolute-potential OM inversion integral from a finite outer radius R to

Numerical method

With E=Phi(r0), the tail is C ∫_R^∞ [d²rho_Q/dPhi²] [dPhi/dr] /sqrt(Phi(r)-E) dr. Use u=ln(r/R), compute potential differences by adaptive quadrature, then integrate u from 0 to 16. The remaining u>16 term is not rigorously enclosed. Leadin

Results

See `tail_results.csv` and `summary.json`. Maximum absolute tail at R/a=10^3 is 1.6739853330457854e-12, and at R/a=10^6 is 1.2006814099823861e-21 in the model's distribution-function normalization. At R/a=10^6 the largest relative discrepan

Limitations

- These are numerical quadrature estimates, **not** rigorous interval-arithmetic error bounds.

- The u>16 remainder was not certified; its asymptotic suppression is expected but not proven as an enclosure.

- This is a tail-only audit, not a second independent complete inversion, not a global positivity proof, and not a stellar-velocity likelihood fit.

- The q≈0.4664668 local tangency and p=1.5 central asymptotic cancellation remain distinct exploratory statements.

- The use of an outer cutoff must be judged together with potential interpolation, quadrature, and central-endpoint errors before any zero-sign certification.

Source excerpt — summary.json (excerpt):

{"model": "Plummer source, p=1.5 tracer, OM; same reference model.py", "rows": 36, "max_abs_tail_at_R1e6a": 1.2006814099823861e-21, "max_abs_tail_at_R1e3a": 1.6739853330457854e-12, "max_asymptotic_relative_discrepancy_at_R1e6a": 6.328055959

Archive item 11: UDCT_v714_Fresh_Local_Rerun_20261030_20261110_124740_169.zip

- SHA-256: `d8d48fcee65a0e4c312eec4bf037d7a187a16948389eb359772bcb5d84abc4ef`; bytes: 8176; nested CRC: pass

- Contained files: `README.md`, `requirements.txt`, `reproduce.py`, `step47_reference.py`, `fresh_df_samples.csv`, `fresh_results.json`, `execution_log.txt`, `archived_audit06_critical_vs_p.csv`, `archived_audit06_optimizer_comparison.csv`, `archived_audit06_curve_derivatives.csv`, `archived_audit06_results.json`

Source excerpt — README.md (excerpt):

UDCT v7.14 Exploratory Findings — p=1.5 Boundary: Reproducibility Supplement

Status: exploratory, not preregistered. **Reserved DOI:** `10.5281/zenodo.23695048` (not registered before publication).

GitHub: https://github.com/wspaik5/UDCT_v7.14_Exploratory_p1.5_Boundary

Run (Python 3.10+)

~~~bash

python -m pip install -r requirements.txt

python reproduce.py

~~~

This generates `fresh_df_samples.csv` and `fresh_results.json` by executing the archived **Step 47 physical kernel** at 36 prespecified local combinations (3 tracer exponents × 3 energy labels × 4 anisotropy settings). This is a **new limit

`source_manifest.csv` records SHA-256 for three unchanged user-supplied primary ZIP archives. They should be uploaded **as separate original ZIPs**, not edited. `archived_audit06_*.csv` and `archived_audit06_results.json` are copied verbati

Scientific limits: p=1.5 marks cancellation of a leading central asymptotic coefficient in the specified model. The finite-energy OM tangency is distinct; finite grids and shared-kernel numerical agreement do not establish continuum non

Source excerpt — fresh_results.json (excerpt):

{"status": "executed fresh local samples", "count": 36, "negative_count": 10, "minimum_f": -0.00016204600413997862, "p_values": [1.499, 1.5, 1.501], "energy_labels": [0.45, 0.474097647, 0.5], "ra_values": ["0.45", "0.466466787396", "0.50",

Archive item 12: UDCT_v714_High_Precision_OM_Crosscheck_REISSUED_20261110_v2_20261110_125130_153.zip

- SHA-256: `25c3008b8f4ac327ededb868a37b52ef81bd932b58a4ea6162db7c796a3cc354`; bytes: 6068; nested CRC: pass

- Contained files: `audit.py`, `comparison.csv`, `summary.json`, `run.log`, `README.md`, `TECHNICAL_NOTE.md`, `requirements.txt`, `SHA256SUMS.txt`

Source excerpt — README.md (excerpt):

UDCT v7.14 — High-Precision Alternative OM Integral Cross-check

Exploratory numerical audit, not a preregistration or global positivity proof.

Reproduce

`python -m pip install -r requirements.txt`

`python audit.py`

The script evaluates three OM anisotropy ratios near the previously recorded local tangency at r/a=0.474097553. It compares the second-derivative inversion integral with an integration-by-parts first-derivative representation using mpmath 3

Caveats

- Precision is not accuracy: inner potential quadrature is fixed at 40 nodes and outer quadrature remains finite.

- Both forms use the same gravity and augmented tracer density and are algebraically related, not independent physical models.

- The outer cutoff is r/a=1e6; no certified tail bound is provided here.

- At the tangency the small value is not a certified zero or a proof of global OM distribution-function positivity.

- Original Steps 01–50 and all preregistration claims remain untouched.

Source excerpt — TECHNICAL_NOTE.md (excerpt):

UDCT v7.14: High-Precision Cross-check of Two OM Inversion Integrals

Scope

Exploratory arithmetic-precision and quadrature comparison for the isolated spherical Plummer gravitational source, p=1.5 tracer, and Osipkov–Merritt augmented density. It does not constitute independent observational validation, a certifie

Methods

The two algebraically equivalent inversion formulas are implemented in `audit.py`: the standard second-derivative kernel and an integration-by-parts representation using the first potential derivative plus an outer boundary term. Both use m

Findings

At the central candidate, 64-node, 55-digit calculations gave 1.7363390385277223e-14 (second derivative) and 1.7362931375604800e-14 (integration by parts), difference 4.5901e-19. With 128 nodes, the difference increased to 2.0308e-18, so mo

The 35- and 55-digit results at 64 nodes agree at the shown precision; this diagnoses arithmetic precision but cannot establish the quadrature error. Because the potential integral uses only 40 nodes and the two expressions share the same m

Next step

Increase *inner* potential quadrature independently (40, 80, 160 nodes) and *outer* quadrature (64, 128, 256) at multiple arithmetic precisions, with explicit finite-tail estimates. Consider a genuinely independent OM inversion algorithm be

Source excerpt — summary.json (excerpt):

{"rows": [{"rr": "0.474097553", "ra": "0.4663667874", "n": 32, "dps": 35, "inner": 40, "direct": "-8.72085029297435185338690-7", "alternative": "-8.72086979983653169135430-7", "integral": "-0.000052369082677611741517017", "boundary": "0.000

Archive item 13: UDCT_v714_Independent_OM_20261109_20261110_124913_839.zip

- SHA-256: `73842527d6331ce5baef1803dbf67051f4cdc94cca109df53a7a5403146d3438`; bytes: 8615; nested CRC: pass

- Contained files: `UDCT_v714_Independent_OM_20261109/`, `UDCT_v714_Independent_OM_20261109/README.md`, `UDCT_v714_Independent_OM_20261109/summary.json`, `UDCT_v714_Independent_OM_20261109/requirements.txt`, `UDCT_v714_Independent_OM_20261109/original_step50.py`, `UDCT_v714_Independent_OM_20261109/run_log.txt`, `UDCT_v714_Independent_OM_20261109/independent_samples.csv`, `UDCT_v714_Independent_OM_20261109/original_step47_reference.py`, `UDCT_v714_Independent_OM_20261109/independent_check.py`

Source excerpt — UDCT_v714_Independent_OM_20261109/README.md (excerpt):

UDCT v7.14 — Step 50 independent OM quadrature check

Exploratory numerical audit, **not** a preregistered confirmatory result.

Run `pip install -r requirements.txt && python independent_check.py`.

This implementation independently reconstructs the Plummer gravitational field, tracer derivatives, OM augmented density and adaptive quadrature. It does **not** import the Step 47 kernel or reuse the original Gauss-Legendre nodes. It check

The 0.46646678739601644 figure is the original Step 50 finite-window tangency, not a universal threshold. The local asymptotic p=1.5 transition is distinct.

`original_step50.py` and `original_step47_reference.py` are archived for provenance only. The new check runs without importing either.

Source excerpt — UDCT_v714_Independent_OM_20261109/summary.json (excerpt):

{"model": "original Step50 spherical algebra, independently implemented scipy adaptive quadrature", "samples": [{"outer_over_a": 1000000.0, "r_energy_over_a": 0.44, "f_iso": 0.0022532277654875137, "B": -0.0004882860430734733, "critical_ra_o

Archive item 14: UDCT_v714_Independent_OM_Crosscheck_FRESH_v2_20261110_124843_115.zip

- SHA-256: `35896e91ed9909c57750a6f5072db368777be2c62f37b6fb3d0e9e7112ac3892`; bytes: 4859; nested CRC: pass

- Contained files: `independent_om_comparison.csv`, `summary.json`, `run_independent.py`, `step47_reference.py`, `requirements.txt`, `README.md`

Source excerpt — README.md (excerpt):

UDCT v7.14 — Independent numerical OM inversion cross-check

Exploratory follow-up; does not amend the archived v7.14 paper.

Method

- Existing Step 47: analytic density-to-potential second derivative and Gauss-Legendre quadrature.

- Independent method: tabulate relative potential from the same gravity model, build cubic spline of augmented density against potential, differentiate spline twice, then integrate over relative potential with Simpson quadrature.

- Fixed model: p=1.5, ra/a=0.466466787396.

Reproduce

`python run_independent.py` from the extracted folder. Requires numpy, scipy, pandas.

Caveats

- Both methods share the gravitational field, so this is not independent physical validation.

- The spline returns -2.354e-10 at the contact point, compared with ~3.76e-15 from Step 47. The discrepancy must NOT be classified as a physical negative without grid convergence and error estimates.

- Other sampled energies show close agreement; finite sampling does not establish global DF positivity.

- The original publication and its locked statements remain unchanged.

Source excerpt — summary.json (excerpt):

{"method": "cubic spline second derivative in potential, independently integrated", "p": 1.5, "ra_over_a": 0.466466787396, "radial_grid": 18000, "energy_points": 7, "limitations": "spline differentiation near tangency is ill-conditioned; fi

Archive item 15: UDCT_v714_Independent_Tangency_Audit_20261109_20261110_124925_712.zip

- SHA-256: `c1b1c5880b498b5aa24a5866072e127ac0d33a84da4f620d8aa1934e140b5a5f`; bytes: 12305; nested CRC: pass

- Contained files: `UDCT_v714_Independent_Tangency_20261109/`, `UDCT_v714_Independent_Tangency_20261109/README.md`, `UDCT_v714_Independent_Tangency_20261109/original_step50_reference.py`, `UDCT_v714_Independent_Tangency_20261109/summary.json`, `UDCT_v714_Independent_Tangency_20261109/optimize_tangency.py`, `UDCT_v714_Independent_Tangency_20261109/requirements.txt`, `UDCT_v714_Independent_Tangency_20261109/optimized_tangencies.csv`, `UDCT_v714_Independent_Tangency_20261109/grid.csv`, `UDCT_v714_Independent_Tangency_20261109/summary_snapshot.json`, `UDCT_v714_Independent_Tangency_20261109/independent_check.py`, `UDCT_v714_Independent_Tangency_20261109/__pycache__/`, `UDCT_v714_Independent_Tang

Source excerpt — UDCT_v714_Independent_Tangency_20261109/README.md (excerpt):

UDCT v7.14 — Independent OM Tangency Re-optimization

Status: exploratory, post-hoc independent numerical audit; NOT preregistered confirmation.

Scope

- Uses the original Step 50 spherical Plummer gravitating source and tracer rho=(1+(r/a)^p)^(-5/p), with p=1.5.

- Computes the OM inversion using a separately written SciPy adaptive quadrature (`independent_check.py`) rather than the original Step 47 Gauss-Legendre kernel.

- Optimizes the local critical ratio over the energy-label interval r/a in [0.25, 0.75] without fixing the Step 50 tangency location.

- Checks finite numerical upper integration cutoffs 1e6 a and 1e7 a.

Run

`python optimize_tangency.py`

Requires Python 3.10+, NumPy and SciPy. Writes `grid.csv`, `optimized_tangencies.csv`, `summary.json`. Reference files from original Step 50 are included for provenance only.

Interpretation

The independent optimization returns r/a ~ 0.474097647 and r_a/a ~ 0.466466787394. The original Step 50 reported r/a ~ 0.474097553 and r_a/a ~ 0.466466787396. This supports local numerical reproducibility, not a globally positive distributi

Limitations: identical physical model assumptions, finite numerical outer radius, limited energy-label search domain; no proof of global DF positivity, dynamical stability, or Hydrus I individual-star observational fit. The local p=1.5

This audit must not be represented as a preregistered test. Future tests must fix hypotheses, ranges, tolerances and decision rules before running new confirmatory calculations.

Source excerpt — UDCT_v714_Independent_Tangency_20261109/summary.json (excerpt):

{"classification": "Exploratory post hoc audit; not preregistered confirmation", "optimization": [{"outer_over_a": 1000000.0, "r_energy_over_a": 0.4740976467364433, "critical_ra_over_a": 0.4664667873940179, "zero_residual": 0.0, "optimizer_

Archive item 16: UDCT_v714_Model_Convention_Gate_REISSUED_20261110_v2_20261110_124959_206.zip

- SHA-256: `1c648c607d7d132070438e3bf055686bfe16e54de4913764e2c208e84a5834e3`; bytes: 11346; nested CRC: pass

- Contained files: `README.md`, `TECHNICAL_NOTE.md`, `audit.py`, `checks.csv`, `source_snapshot.json`, `source_model.py`, `reference_direct_tangency.zip`, `run.log`, `SHA256SUMS.txt`

Source excerpt — README.md (excerpt):

UDCT v7.14 — Model-Convention Gate

Offline source audit (not a new physics test or preregistration).

Run `python audit.py` with Python 3 standard library only. It extracts the archived `model.py`, checks presence of specific model conventions and records SHA-256. These are structural/source checks, not mathematical proofs. See TECHNICAL_NO

Source excerpt — TECHNICAL_NOTE.md (excerpt):

UDCT v7.14 — Model-Convention Source Gate (Exploratory Audit)

Status: Pre-preregistration source audit only. No new confirmatory physics experiment, no certification of the DF sign, and no new observational fit.

Inspected source

`model.py` extracted from the archived `UDCT_v714_Direct_Tangency_Refinement_20261109.zip`, SHA-256 recorded in `source_snapshot.json`. This package embeds the original ZIP for offline reproducibility.

Source-derived model definitions

The code sets `G=4.30091e-3`, `a=53.3`, `M=32730.`, with acceleration scale `a0=1.2e-10*3.085677581491367e16/1e6`. The gravitational Newtonian acceleration is `n=G*M*r/(r*r+a*a)**1.5` (Plummer mass source). The effective acceleration is `g=

The kernel is `(rho_aug_second_derivative*g-rho_aug_first_derivative*gprime)/g**3`. The direct-potential routine computes the potential difference by adaptive integration along logarithmic radius, followed by Gauss–Legendre quadrature over

Mathematical caution

For a finite total Plummer source and the stated MOND-like acceleration, g asymptotically scales as 1/r in the deep-MOND outer region, so the potential difference grows logarithmically as r approaches infinity. A standard finite-escape-ener

Protocol gates still open

1. Derive and verify the OM inversion and its boundary term for the log-diverging outer potential, including normalization and energy convention.

2. Define a finite dimensionless energy domain and show the radius-to-energy mapping.

3. Bound truncation, interpolation, quadrature and floating-point errors if claiming certified signs. Ot

Source excerpt — source_snapshot.json (excerpt):

{"source_zip": "reference_direct_tangency.zip", "model_py_sha256": "ba6e38fd19f2dd8dbfacf2d00f833f38134504eb5030f2c4dee2e69b44934ec4", "checks": {"model.py in archived ZIP": true, "gravity defined": true, "kernel defined": true, "direct pot

Archive item 17: UDCT_v714_Nested_Quadrature_Gate_REPRO_20261110_20261110_125137_017.zip

- SHA-256: `dce2eee8c3bb772e7c89920cf8e8fe2a526058a6e97c5fabb3cf9038a4853465`; bytes: 7788; nested CRC: pass

- Contained files: `README.md`, `SHA256SUMS.txt`, `TECHNICAL_NOTE.md`, `audit_nested.py`, `nested_grid.csv`, `reference_audit.py`, `requirements.txt`, `run.log`, `summary.json`

Source excerpt — README.md (excerpt):

UDCT v7.14 Nested Quadrature Gate — exploratory

Run: `python audit_nested.py` (requires Python 3, numpy, mpmath).

Outputs: `nested_grid.csv` (27 rows), `summary.json`, `run.log`. `reference_audit.py` is an unmodified copy of the prior high-precision OM cross-check audit, used as the calculation engine. `audit_nested.py` drives the 3 x 3 x 3 grid. No ne

This package is a prospective preparation audit, not a preregistered test. See TECHNICAL_NOTE.md for exact scope and caveats.

Source excerpt — TECHNICAL_NOTE.md (excerpt):

UDCT v7.14 — Nested Quadrature Convergence Gate (Exploratory)

Date: 2026-11-10. Status: exploratory numerical audit, NOT preregistered confirmation.

Model and methods

The audit reuses the reference `audit.py` from the earlier high-precision OM cross-check without modifying its mathematical conventions: isolated spherical Plummer source, p=1.5 tracer, Osipkov–Merritt augmented density, MOND-like algebraic

The tangency-energy proxy is fixed at r/a=0.474097553. The three anisotropy values are r_a/a = 0.4663667874, 0.46646678739601644, 0.4665667874. Total 27 calculations.

Key results

At 160 inner / 256 outer nodes, the direct and alternative evaluations are:

| r_a/a | direct | alternative | absolute discrepancy |

|---|---|---|---|

| 0.4663667874 | -8.72087036102638178302 84e-7 | -8.72087036102736913326 99e-7 | 9.87e-20 |

| 0.46646678739601644 | 1.736318964594618977236e-14 | 1.736309120993447256422 7e-14 | 9.84e-20 |

| 0.4665667874 | 8.715264529530616350724 7e-7 | 8.715264529529634978809 7e-7 | 9.81e-20 |

For the tangency candidate, the largest change from the 160-inner/256-outer reference among inner nodes at fixed 256 outer nodes is 2.28e-20; among outer nodes at fixed 160 inner nodes it is 3.71e-18. This indicates the outer OM quadrature

Limitations and next gate

The expressions share density, gravity, potential, cutoff, and much code. There is no rigorous interval-arithmetic enclosure, no indep

Source excerpt — summary.json (excerpt):

{"0.4663667874": {"reference_inner160_outer256_direct": "-8.72087036102638178302 84e-7", "reference_inner160_outer256_alternative": "-8.72087036102736913326 99e-7", "max_expression_discrepancy": "2.2956107778039893E-18", "max_inner_change_at_

Archive item 18: UDCT_v714_Newtonian_OM_Control_REISSUED_20261029_20261110_124620_851.zip

- SHA-256: `39a150b4ac37db0ac882a4abadacd68d6ae6e51260e45339153c31de4c41038f`; bytes: 5313; nested CRC: pass

- Contained files: `UDCT_v714_Newtonian_OM_Control/`, `UDCT_v714_Newtonian_OM_Control/README.md`, `UDCT_v714_Newtonian_OM_Control/results.csv`, `UDCT_v714_Newtonian_OM_Control/newtonian_plummer_om_control.py`, `UDCT_v714_Newtonian_OM_Control/results.json`

Source excerpt — UDCT_v714_Newtonian_OM_Control/README.md (excerpt):

UDCT v7.14 — supplementary Newtonian Plummer–OM control

This is a **supplementary control**, not a replacement for the original v7.14 protocol or a preregistered QUMOND result.

Scope and assumptions

Self-consistent Newtonian Plummer mass and tracer profile with potential `Psi=GM/sqrt(r^2+a^2)`; OM augmented density `rho_Q=(1+r^2/r_a^2)rho`. Units `G=M=a=1`; `Q` is relative OM energy with `0<Q<1`. These assumptions matter: a Plummer tra

Analytic calculation

`rho = 3/(4*pi) * Psi^5`, `r^2=Psi^(-2)-1`, so

`rho_Q = 3/(4*pi) * [ (1-1/s)*Psi^5 + (1/s)*Psi^3 ]`, with `s=(r_a/a)^2`.

Eddington/OM inversion gives

`f(Q) = [3/(4*pi*sqrt(8)*pi^2)] * Q^(3/2) * [ 8/s + (128/7)*(1-1/s)*Q^2 ]`.

For `s<1`, the bracket is minimized as `Q->1`; nonnegativity over the entire allowed energy range requires `s>=9/16`, or **r_a/a >= 3/4**. At exactly 3/4, the limiting central high-energy value vanishes; for all interior `0<Q<1` it is posit

Reproduction

Requires Python 3.9+ and SciPy (`python -m pip install scipy`). Run `python newtonian_plummer_om_control.py` in this directory. Generates `results.csv` and `results.json`; cross-checks the analytic expression against a separately evaluated

Research interpretation

Source excerpt — UDCT_v714_Newtonian_OM_Control/results.json (excerpt):

{"model": "Newtonian self-consistent isotropic-mass Plummer potential with OM tracer DF", "units": "G=M=a=1", "exact_threshold_ra_over_a": 0.75, "worst_absolute_quadrature_difference": 2.1316282072803006e-14, "pass_quadrature": true, "note"

Archive item 19: UDCT_v714_OM_Alternative_Integral_Audit_20261110_20261110_125123_536.zip

- SHA-256: `6fb8c5b772fd59683e1651a9bdbe2089b2342d465a4140fdab6f7bcb44e43c61`; bytes: 8052; nested CRC: pass

- Contained files: `README.md`, `SHA256SUMS.txt`, `TECHNICAL_NOTE.md`, `audit.py`, `comparison.csv`, `model.py`, `requirements.txt`, `run.log`, `summary.json`

Source excerpt — README.md (excerpt):

UDCT v7.14 Alternative OM Integral Audit

Exploratory continuation of the v7.14 series; NOT the formal preregistration.

Run: `python audit.py` (requires numpy and scipy). The script writes `comparison.csv` and `summary.json`. `model.py` is copied unchanged from the earlier FD step-size audit to preserve model conventions. `TECHNICAL_NOTE.md` contains the der

CAUTION: At the candidate tangency the alternative integral suffers catastrophic cancellation. It is not evidence of an exact zero, global phase-space positivity, or Hydrus I stellar-velocity fitting.

Source excerpt — TECHNICAL_NOTE.md (excerpt):

UDCT v7.14 — Alternative OM Integral Representation Audit

Status: Exploratory numerical cross-check, not preregistration and not proof of global positivity.

Model

Uses the unchanged `model.py` from the prior FD step-size package: Plummer spherical gravitational source, p=1.5 tracer, OM augmented density, isolated MOND-like acceleration. The reference radial point is a label for Q = Phi(r0)-Phi(0), an

Mathematically distinct integral representation

Write D(Phi) = d rho_Q/d Phi and Delta = Phi - Q, where Q = Phi(r0). With C=1/(sqrt(8) pi^2), the original truncated integral is

f_R(Q) = C * integral_Q^PhiR D'(Phi)/sqrt(Phi-Q) dPhi.

Integrating by parts after subtracting D(Q) gives the EXACT finite-upper-limit identity

f_R(Q) = (C/2) * integral_Q^PhiR [D(Phi)-D(Q)]/(Phi-Q)^(3/2) dPhi

+ C * [D(PhiR)-D(Q)]/sqrt(PhiR-Q).

The explicit upper-boundary term is essential. The subtraction makes the lower endpoint integrable. We evaluate Phi(r)-Phi(r0) by adaptive quadrature at each Gaussian node, and D(r) from the analytic first radial derivative divided by g(r).

Findings

27 calculations: three anisotropy radii, three r/a labels, three quadrature orders. Near r/a=0.474097553 and ra/a=0.46646678739601644, the second-derivative integral is ~1.736e-14, while the alternative is ~1.67e-14 (64 nodes), ~1.42e-14 (1

Source excerpt — summary.json (excerpt):

{"n": 27, "max_abs_difference": 4.2364955944382277e-14, "rows": [{"rr": 0.2, "ra": 0.4663667874, "n": 64, "outer": 1000000.0, "direct": 0.0015016951678751532, "alternative": 0.0015016951678777535, "alternative_integral": 0.00141438187891750

Archive item 20: UDCT_v714_OM_Derivative_Crosscheck_REISSUED_20261110_v2_20261110_125104_335.zip

- SHA-256: `6aa980f9f9f713e37b84d496c1eaf57aaddb3bf0bf53107d90bf06bafe51be7b`; bytes: 7024; nested CRC: pass

- Contained files: `model.py`, `audit.py`, `comparison.csv`, `summary.json`, `README.md`, `TECHNICAL_NOTE.md`, `requirements.txt`, `run.log`, `SHA256SUMS.txt`

Source excerpt — README.md (excerpt):

UDCT v7.14 — Numerical-Derivative Cross-check (exploratory)

Run `python audit.py` (requires numpy and scipy). This package compares an analytic second derivative of the OM augmented tracer density against a separately coded finite difference of its first derivative. The gravitational potential is in

Source excerpt — TECHNICAL_NOTE.md (excerpt):

UDCT v7.14: OM Numerical-Derivative Cross-check

Scope

Exploratory audit of the original isolated Plummer source, p=1.5 tracer and OM augmented density. Six r/a points (0.1, 0.3, 0.474097553, 0.7, 1, 3) and three anisotropy ratios (0.4663667874, 0.46646678739601644, 0.4665667874) were evaluated

Method

Path A evaluates the analytic second derivative of augmented density with respect to potential, as in the source model. Path B differentiates the analytic first derivative numerically using centered log-radius differences h = 1e-3, 3e-4, 1e

Findings

- 18 samples; maximum absolute discrepancy at h=1e-4: 6.285321e-12.

- At r/a=0.474097553, ra/a=0.46646678739601644: analytic 1.736418754e-14; numerical derivative h=1e-4 5.190736785e-12.

- This ~5e-12 difference dominates the ~1.7e-14 candidate value. No robust zero-sign determination is possible from this cross-check alone.

- Away from the tangency, the signs match for the three sampled ra/a values at r/a=0.474097553.

Limitations and next action

Finite difference truncation and cancellation error remain; a truly independent inversion, interval arithmetic and a predefined finite energy domain remain outstanding. Do not interpret this as global distribution-function positivity, a uni

Source excerpt — summary.json (excerpt):

{"n_samples": 18, "max_abs_difference": 6.2853212218216825e-12, "max_relative_discrepancy": 297.9334670591956, "tangency_row": {"r_over_a": 0.474097553, "ra_over_a": 0.46646678739601644, "analytic": 1.7364187544592037e-14, "fd_h1e3": 5.1735

Archive item 21: UDCT_v714_OM_Gate_Download_Fix_02_20261110_124731_146.zip

- SHA-256: `bdad29e59af54f5de6a0f2014ed5013730cd1e01bebd894135020aba27c5c7dd`; bytes: 4186; nested CRC: pass

- Contained files: `run.py`, `README.md`, `necessary_gate.csv`, `summary.json`

Source excerpt — README.md (excerpt):

UDCT v7.14 Protocol 01 — OM augmented-density necessary-condition audit

Exploratory / not preregistered / not an OM distribution-function inversion.

Execute

`python run.py` (Python 3.9+, standard library only).

Outputs: `necessary_gate.csv` and `summary.json`.

Mathematical condition

For a monotone spherical relative potential Ψ(r) decreasing outward, a nonnegative Osipkov–Merritt DF implies nonnegative derivative $d\rho_Q/d\Psi$, hence $d\rho_Q/dr \le 0$, where $\rho_Q=(1+r^2/r_a^2)\rho(r)$. This is **necessary, not sufficient** for $f(Q)\ge 0$.

Using x=r/a, b=r_a/a and generalized tracer ρ=(1+x^p)^(-5/p):

`d ln ρ_Q/dx = 2x/(b²+x²) - 5 x^(p-1)/(1+x^p)`.

For $0<p<2$, the exact necessary bound is $b^2 \ge \max_{x>0}[(2/5)x^{2-p} - (3/5)x^2]$, whose maximum occurs at $x^p=(2-p)/3$. At p=2 the exact necessary bound is $b\ge\sqrt{2/5}\approx 0.632455532$; for p>2, no finite b passes the central condition in this par

For the self-consistent Newtonian Plummer OM model, the stronger known full-DF boundary is b≥0.75. A necessary gate at 0.63246 does not contradict that.

Source excerpt — summary.json (excerpt):

{"status": "exploratory necessary-condition only", "density": "rho=(1+x^p)^(-5/p)", "augmented_density": "(1+x^2/b^2)rho", "p2_exact_necessary_b_min": 0.6324555320336759, "p2_exact_full_OM_b_min_from_prior_control": 0.75, "candidate_b": 0.4

Archive item 22: UDCT_v714_OM_Inversion_REISSUED_20261029_v2_20261110_124537_978.zip

- SHA-256: `a5621db977cc63388b0cabce19607ccae4573ee800ac3f58de96ea3a45fb0855`; bytes: 15014; nested CRC: pass

- Contained files: `om_independent.py`, `df_scan.csv`, `summary.json`, `README.md`

Source excerpt — README.md (excerpt):

UDCT v7.14 Protocol 01 — independent OM inversion (exploratory)

Run: `pip install numpy scipy && python om_independent.py`.

This script uses an **unbounded absolute-potential energy convention** for the isolated spherical simple-MOND Plummer gravitational field, rather than a finite escape-potential convention. Define E=Phi(r)-Phi(0), with Phi'(r)=g(r)>0, and E

Preliminary numeric findings: On a 44-point energy/radius grid (r_E/a from 0.005 to 30), q=r_a/a=0.46 gives negative f at two points; q=0.4664667874 gives no negative samples. This does NOT prove that 0.4664667874 is the global critical

Caveats: The script treats the Plummer profile as the gravitating source and separately uses a p=1.5 tracer density; this is not a direct Hydrus I star-level likelihood fit. No external field, dynamical stability, or observational valid

Source excerpt — summary.json (excerpt):

{"method": "Abel inversion on unbounded absolute potential, log-r integration with algebraic endpoint weight", "q_scanned": [0.406320517, 0.42, 0.44, 0.46, 0.4664667874, 0.47, 0.5, 0.55, 0.65, 0.8, 1.0], "x_range": [0.005, 30.0], "samples_p

Archive item 23: UDCT_v714_OM_Outer_Boundary_Audit_REISSUED_20261110_v2_20261110_125005_485.zip

- SHA-256: `fe538452418300d5b81d9c2bdf5a753b6e2de4a66d812e6088263a56d68f602e`; bytes: 6753; nested CRC: pass

- Contained files: `README.md`, `SHA256SUMS.txt`, `TECHNICAL_NOTE.md`, `asymptotic_table.csv`, `audit.py`, `reference_model.py`, `run.log`, `summary.json`

Source excerpt — README.md (excerpt):

Reproducible exploratory OM outer-boundary check

Run `python audit.py` (Python 3 with SciPy installed). The script writes `asymptotic_table.csv` and `summary.json`; the included `run.log` is the recorded output. Units and constants follow the archived v7.14 direct-tangency source. The aud

Source reference: `UDCT_v714_Direct_Tangency_Refinement_20261109.zip`, `model.py` (model conventions). Source snapshot included in this ZIP for traceability.

Source excerpt — TECHNICAL_NOTE.md (excerpt):

UDCT v7.14 — OM Outer-Boundary Asymptotic Audit

Status: exploratory analytical/numerical consistency check, not preregistered confirmation.

Scope

We inspect the isolated spherical Plummer gravitational source, the p=1.5 tracer density, and Osipkov–Merritt augmentation at q=r_a/a=0.46646678739601644. The MOND acceleration convention is taken directly from the archived direct-tangency

Define x=r/a, rho(x)=(1+x^p)^(-5/p), and rho_Q(x)=(1+x^2/q^2)rho(x). For x→∞, rho(x)~x^-5 and rho_Q(x)~x^-3/q^2. The isolated field satisfies dPhi/dx ~ V2/x, where V2=sqrt(G M a0). Therefore:

- x^3 rho_Q → 1/q^2;

- x^3 d(rho_Q)/dPhi → -3/(q^2 V2);

- x^3 d²(rho_Q)/dPhi² → 9/(q^2 V2²).

Consequently d(rho_Q)/dPhi→0 as Phi→∞. For a logarithmically unbounded potential, the usual *finite escape-energy* boundary convention must not be inserted without rederivation. With the standard isotropic/OM Abel convention using absolute,

Computed check

`audit.py` evaluates the analytic expressions at x=10,100,...,10^6. The x=10^6 relative differences from the four predicted limits are recorded in `summary.json`. These small deviations demonstrate agreement with leading asymptotics for the

Unresolved issues

1. A certified or conservative bound on the finite outer-radius tail integral remains absent.

2. The central-endpoint regularity, full energy-domain specification, and a genuinely independent OM DF inversion remain to be audited.

3. Neither global DF positivity nor a universal physical threshold is established.

Source excerpt — summary.json (excerpt):

{"model": "isolated spherical Plummer mass; p=1.5 tracer; OM q=ra/a; simple MOND acceleration interpolation", "V2_asymptotic_km2_s2": 22.83069202336987, "q": 0.46646678739601644, "predicted_limits": {"x3_rho_Q": 4.595772747469026, "x3_d_rho

Archive item 24: UDCT_v714_Positive_Energy_Window_Gate_REISSUED_20261110_v2_20261110_125036_708.zip

- SHA-256: `6b0f580918a3d4d8e019401ea35bab529ecabf1e4449d4c57f4ca44849db13f2`; bytes: 7292; nested CRC: pass

- Contained files: `README.md`, `SHA256SUMS.txt`, `TECHNICAL_NOTE.md`, `audit.py`, `requirements.txt`, `run.log`, `summary.json`, `window_results.csv`

Source excerpt — README.md (excerpt):

UDCT v7.14 — Positive-Energy Window Gate

Exploratory follow-up to Central Regularity Gate, 2026-11-10.

Run with Python 3, NumPy and SciPy:

~~~bash

python -m pip install -r requirements.txt

python audit.py

~~~

The script regenerates `window_results.csv` and `summary.json`. Read `TECHNICAL_NOTE.md` for the physical conventions, method, sign screen and limits. `run.log` is the original execution output. `SHA256SUMS.txt` records the included files'

Source excerpt — TECHNICAL_NOTE.md (excerpt):

UDCT v7.14 — Positive-Energy Window and Central-Stability Gate

Status: exploratory numerical audit; NOT preregistration or proof of global phase-space admissibility.

Definition and provenance

This follow-up retains the isolated spherical Plummer gravitational source, the specified MOND interpolation `g = g_N/2 + sqrt(g_N²/4 + a0*g_N)`, the tracer `rho=(1+(r/a)^1.5)^(-5/1.5)`, and the OM augmented tracer `(1+r²/r_a²)rho`, with fi

For an isolated MOND potential increasing logarithmically outward, the energy label used here is `Q=Phi(r0)-Phi(0)` with `r0>0`, hence `Q>0`. The code parameterizes energies by turning radii `r0/a`, not by a uniformly spaced Q grid. The for

Method

At each of 21 preselected `r0/a` values, the OM kernel `d²rho_Q/dPhi²` is integrated from `r0` to `10^6 a` using a square-root endpoint substitution and 64-point or 128-point Gauss-Legendre quadrature. Every potential difference `Phi(r)-Phi

Results

- Near-centre `1e-5 <= r0/a <= 1e-2`: 7 positive, 0 negative, 0 indeterminate.

- Intermediate `0.03 <= r0/a <= 3`: 10 positive, 0 negative, 1 indeterminate.

- Outer `10 <= r0/a <= 100`: 3 positive, 0 negative, 0 indeterminate.

- The single indeterminate sample is `r0/a=0.474097553`, with `f_128=1.7358937e-14`.

- Maximum observed 64-to-128 order difference is `1.4856701877e-7`, at `r0/a=1e-5`; this is not a rigorous error bound.

Interpretation and constraints

The sampled finite-energy points are consistent with nonnegative OM inversion under the chosen numerical screen, but do **not

Source excerpt — summary.json (excerpt):

{"model": "isolated spherical MOND interpolation, Plummer gravity source, p=1.5 tracer, OM augmented density", "ra_over_a": 0.46646678739601644, "outer_radius_over_a": 1000000.0, "quadrature_orders": [64, 128], "n_samples": 21, "windows": {

Archive item 25: UDCT_v714_Precision_OM_Audit_NEW_LINK_20261030_20261110_124813_703.zip

- SHA-256: `f01e720065ff5be432cedccd7aa2d8f7b0f5c0578081a13a4b8672de81dc6113`; bytes: 9414; nested CRC: pass

- Contained files: `step47_reference.py`, `run_precision.py`, `precision_scan.csv`, `precision_results.json`, `README_PRECISION.md`, `run_extended.py`, `extended_scan.csv`, `extended_results.json`, `requirements.txt`

Source excerpt — precision_results.json (excerpt):

[{"p": 1.5, "ra": 0.466466787396, "nodes": 192, "samples": 23, "minimum": -8.190515268818265e-17, "at": 0.474097647, "negative_below_minus_1e_10": 0}, {"p": 1.5, "ra": 0.466466787396, "nodes": 384, "samples": 23, "minimum": 7.18397404171170

Archive item 26: UDCT_v714_Prospective_Protocol_Gate_DRAFT_20261109_20261110_124951_662.zip

- SHA-256: `d5fd940ec150db78f245d85d58fdcf091587215298ec3499873a3aa5dbe9e23c`; bytes: 4659; nested CRC: pass

- Contained files: `SHA256SUMS.txt`, `PROTOCOL_GATE_DRAFT.md`, `gate_matrix.csv`, `run.log`, `verify.py`, `source_snapshot.json`, `README.md`

Source excerpt — README.md (excerpt):

UDCT v7.14 protocol release-gate draft. Run `python verify.py` to verify the preceding audit summary and 12-row matrix. No new physics computation.

Source excerpt — source_snapshot.json (excerpt):

{"original_step50": {"nested_archive": "UDCT_v7_14_Step50_OM_Tangency_REDOWNLOAD_READY_20261026_20261026_023159_397.zip", "p": 1.5, "critical_ra_over_a": 0.46646678739601644, "tangent_r_over_a": 0.47409755283222904, "declared_domain": "r/a

Archive item 27: UDCT_v714_Protocol_Lock_Readiness_20261110_20261110_125142_622.zip

- SHA-256: `2eaeacf0a8ae4e68b0bbf1ef3b656232cc2815ca11e746bf9ef6a0bdd1ef347c`; bytes: 4237; nested CRC: pass

- Contained files: `PROTOCOL_CANDIDATE.md`, `README.md`, `SHA256SUMS.txt`, `audit.py`, `reference_checks.csv`, `summary.json`

Source excerpt — README.md (excerpt):

UDCT v7.14 Protocol Lock Readiness

Run `python audit.py` after placing `UDCT_v714_Nested_Quadrature_Gate_REPRO_20261110.zip` at the path specified in audit.py, or change the `source` path to its local location. Python standard library only. `reference_checks.csv` and `summar

Source excerpt — summary.json (excerpt):

{"source_zip_sha256": "dce2eee8c3bb772e7c89920cf8e8fe2a526058a6e97c5fabb3cf9038a4853465", "checks": {"archive_crc_ok": true, "rows_27": true, "two_expressions": true, "cutoff_fixed_1e6": true, "potential_inner_gauss": true, "outer_om_gauss"

Archive item 28: UDCT_v714_Protocol01_Asymptotic_Notes_20261110_124440_189.md

- SHA-256: `192e5d6be742c3b47a14a9755ad878423d5a65fe6b0072153e8681ea0d472470`; bytes: 2146; nested CRC: N/A

- Contained files:

Source excerpt — UDCT_v714_Protocol01_Asymptotic_Notes_20261110_124440_189.md (excerpt):

UDCT v7.14 Protocol 01 — Independent asymptotic derivation (pre-lock)

Status: Symbolic verification of an already explored asymptotic result; NOT a new prospective numerical experiment. The companion Python script uses SymPy and does not import archived Step 40 or Step 47 code.

Model and expansion

For the isolated spherical simple-QUMOND Plummer source, define `k=GM/a^3`, `h=sqrt(a0*k)` and central outward potential difference `Psi(r)=A*r^(3/2)+B*r^2+O(r^(5/2))`, with `A=2h/3>0`, `B=k/4>0`. The tracer profile is `rho_p(r)=[1+(r/a)^p]

Introduce `u=(Psi/A)^(2/3)`. Series reversion gives `r=u*[1-(2B/(3A))*u^(1/2)+O(u)]`. Therefore

`rho_p(Psi)=1-C*(Psi/A)^(2p/3)+(2p*C*B/(3A))*(Psi/A)^((2p+1)/3)+...`,

subject to accounting for competing tracer terms from `rho_p(r)=1-C*r^p+O(r^(2p))`. At `p=3/2`, these competing terms begin at `Psi^2` and thus occur after the displayed `Psi^(4/3)` correction.

At `p=3/2` the leading `Psi^1` term has zero second derivative, whereas the next term is `+(C*B/A^(7/3))*Psi^(4/3)` and

`rho''(Psi)=(4/9)*(C*B/A^(7/3))*Psi^(-2/3)+... > 0`.

This reproduces the **local asymptotic sign and exponent** already explored in Step 40, via explicit series reversion. It does not prove positivity of the full DF, stability, or an observational fit. The coefficient of the Abel-inverted lea

What is and is not independent

Archive item 29: UDCT_v714_Protocol01_Continuation_20261029_20261110_124643_378.zip

- SHA-256: `d7f3d7c96f364cade60c25d53bd4f3e4329b70cd1f625d48a23009495bbcd0f5`; bytes: 19759; nested CRC: pass

- Contained files: `README.md`, `BASELINE_ASYMPTOTIC_VERIFICATION.zip`, `OM_DESIGN_GATE.zip`, `NEWTONIAN_PLUMMER_OM_CONTROL.zip`

Source excerpt — README.md (excerpt):

UDCT v7.14 Protocol 01 — Continuation bundle (2026-10-29)

This bundle preserves prior independently prepared packages. No new confirmatory OM inversion has been executed in assembling this bundle.

Contents

- BASELINE_ASYMPTOTIC_VERIFICATION.zip: independent symbolic/finite-radius check and scripts

- OM_DESIGN_GATE.zip: critical outer-potential convention issue and prospective test plan

- NEWTONIAN_PLUMMER_OM_CONTROL.zip: Newtonian self-consistent Plummer OM control

Required next decision before prospective OM inversion

Freeze a mathematically justified finite energy/potential convention (outer truncation or explicit external-field model) and all numerical tolerances. A spherical isolated MOND field has logarithmically divergent potential at infinity, so c

The exploratory 0.4664667874 QUMOND candidate is not independently confirmed by this bundle. Newtonian 0.75 applies to the self-consistent Newtonian Plummer model only.

Do not reinterpret exploratory results as preregistered results.

Archive item 30: UDCT_v714_Protocol01_CrossQuadrature_20261029_20261110_124702_757.zip

- SHA-256: `253a9e521d9cef2a3a1b802f545e181d536d817ec8e6d7a9b7cba0deb2d9f3ac`; bytes: 10097; nested CRC: pass

- Contained files: `UDCT_v714_Protocol01_CrossQuadrature_20261029/`,
`UDCT_v714_Protocol01_CrossQuadrature_20261029/cross_check.csv`,
`UDCT_v714_Protocol01_CrossQuadrature_20261029/cross_quadrature.py`,
`UDCT_v714_Protocol01_CrossQuadrature_20261029/summary.json`,
`UDCT_v714_Protocol01_CrossQuadrature_20261029/__pycache__/`,
`UDCT_v714_Protocol01_CrossQuadrature_20261029/__pycache__/baseline_reference.cpython-313.pyc`,
`UDCT_v714_Protocol01_CrossQuadrature_20261029/baseline_reference.py`,
`UDCT_v714_Protocol01_CrossQuadrature_20261029/README.md`

Source excerpt — UDCT_v714_Protocol01_CrossQuadrature_20261029/README.md (excerpt):

UDCT v7.14 — Protocol 01 exploratory quadrature cross-check

Run `pip install numpy scipy` then `python cross_quadrature.py` in this folder.

This reproduces a *limited numerical cross-check* of the OM inversion used in the prior exploratory package. The new method changes the integration variable to t=sqrt(log(r/r_E)) and uses ordinary adaptive quadrature; the original uses loga

Inputs: isolated spherical simple-MOND field sourced by a Plummer profile, separate p=1.5 tracer, unbounded absolute potential convention, truncated r/a=10^6. Results are 24 sampled points, not global f(Q) positivity or a precise threshold

Source excerpt — UDCT_v714_Protocol01_CrossQuadrature_20261029/summary.json (excerpt):

{"samples": 24, "max_absolute_difference": 3.4751384062059465e-11, "negative_substitution": 1, "negative_weighted": 1, "scope": "Exploratory, shared potential/density derivatives, distinct quadrature variable; not fully independent model im

Archive item 31: UDCT_v714_Protocol01_DRAFT_20261110_124425_770.json

- SHA-256: `61d3a8af5ab98dedfe3cfa9f362f822a3a617fa4c652814cfd8965f6d3492368`; bytes: 1258; nested CRC: N/A

- Contained files:

Source excerpt — UDCT_v714_Protocol01_DRAFT_20261110_124425_770.json (excerpt):

{

"status": "DRAFT_NOT_FROZEN_NOT_EXECUTED",

"protocol": "UDCT_v7.14_Protocol01",

"archive_sha256": {

"7.14 Exploratory audit 1~6.zip": "6980b333823abba340403d6d921b50805291e42b5f8b810ab003bce19ef44614",

"7.14 Hessian 1~25.zip": "24e646cb99f0b0cd0f997e9afba68f2be3d3a01f68ad7b222e3d943cb66924ff",

"7.14 Hessian 26~50.zip": "14da6463e2dc1b9af6726c448517aa6013709ae2e967a3b5cb8d7ae985292f58"

},

"historical_benchmarks_not_prospective": {

"step50_p": 1.5,

"step50_local_ra_over_a": 0.46646678739601644,

"step50_energy_label_r_over_a": 0.47409755283222904,

"step50_search_interval": [

0.25,

0.75

],

"audit03_p": [

1.49,

1.495,

1.499,

1.5,

1.501,

Archive item 32: UDCT_v714_Protocol01_Gravity_Gate_20261029_REISSUED_20261110_124724_009.zip

- SHA-256: `6eaed323b619dec9ea8ad20681fd397e9a724169a739e9e1da7906b06c94cb08`; bytes: 5030; nested CRC: pass

- Contained files: `run.py`, `README.md`, `summary.json`, `requirements.txt`, `finite_boundary_potential.csv`, `gravity_crosscheck.csv`

Source excerpt — README.md (excerpt):

UDCT v7.14 — Protocol 01: independent spherical gravity field gate

Exploratory cross-check, **not preregistered confirmation**. Reconstruct Plummer enclosed mass from a 1D density integral, independently compare with closed-form mass, then evaluate spherical simple-MOND acceleration. Compare finite-radius

Run

`python -m pip install -r requirements.txt`

`python run.py`

Outputs

`gravity_crosscheck.csv`, `finite_boundary_potential.csv`, `summary.json`.

Limits

- In spherical symmetry the algebraic MOND relation is the spherical QUMOND acceleration for the adopted interpolation. This does not test nonspherical QUMOND or GQUMOND Hessian effects.

- `Phi(R)-Phi(r)` depends on the selected finite outer radius R. A finite reference radius is **not** a self-consistent physical truncation of a Plummer distribution. An OM inversion requires a properly specified global potential and tracer

- No DF positivity test or independent reproduction of the exploratory r_a/a ~ 0.4665 threshold is claimed.

- p=1.5 is a separate local asymptotic tracer-density statement and is not evaluated in this package.

- This test compares independent mass-integration routes but still shares the spherical acceleration law; it is not a wholly independent field-equation solver.

Source excerpt — summary.json (excerpt):

{"status": "exploratory independent gravity field crosscheck; no OM inversion", "max_relative_mass_difference": 3.950334515671548e-16, "max_relative_acceleration_difference": 3.73291938620993e-16, "max_relative_potential_quadrature_differen

Archive item 33: UDCT_v714_Protocol01_Independent_Asymptotic_20261110_124442_637.py

- SHA-256: `3d62cf7b512cbe5b44eab73a369d8c86e6f260f5e2de0ccb30a73b38b6e0d31f`; bytes: 1190; nested CRC: N/A

- Contained files:

Source excerpt — UDCT_v714_Protocol01_Independent_Asymptotic_20261110_124442_637.py (excerpt):

"""UDCT v7.14 Protocol 01: symbolic-only independent local asymptotic audit.

No finite-energy inversion, prospective parameter sweep, or post-registration tests.

Requires sympy. Execute: python UDCT_v714_Protocol01_Independent_Asymptotic.py

"""

import sympy as s

p,C,A,B,Psi=s.symbols('p C A B Psi',positive=True)

alpha=2*p/3

Potential Psi=A*r^(3/2)+B*r^2+O(r^(5/2)). Let u=(Psi/A)^(2/3).

Series inversion r=u*(1-(2/3)*(B/A)*u^(1/2)+O(u)).

u=s.symbols('u',positive=True)

r=u*(1-s.Rational(2,3)*(B/A)*s.sqrt(u))

rho=1-C*r^p+... ; retain terms linear in B.

leading=-C*u**p

correction=s.expand(-C*u**p*(-p*s.Rational(2,3)*(B/A)*s.sqrt(u)))

Re-express u=(Psi/A)^(2/3).

sublead=s.simplify(correction.subs(u,(Psi/A)**s.Rational(2,3)))

critical=s.simplify(sublead.subs(p,s.Rational(3,2)))

second=s.simplify(s.diff(critical,Psi,2))

assert s.simplify(second/(s.Rational(4,9)*C*B/A**s.Rational(7,3)*Psi**(-s.Rational(2,3))))==1

print('General subleading term:',sublead)

print('At p=3/2:',critical)

print('Second derivative at p=3/2:',second)

Archive item 34: UDCT_v714_Protocol01_NextStage_DesignGate_20261029_20261110_124515_076.zip

- SHA-256: `79324efed04d9c23a0d51a8f4fd0ac4c9aece335abfb2cfe1162bfe8fd9bd69f`; bytes: 10553; nested CRC: pass

- Contained files: `UDCT_v714_Protocol01_NextStage/`, `UDCT_v714_Protocol01_NextStage/PROTOCOL01_NEXT_STAGE_DESIGN.md`, `UDCT_v714_Protocol01_NextStage/BASELINE_VERIFICATION_ARCHIVE.zip`

This ledger entry contains no directly quoted technical note in the prior reconstruction; consult the companion archive for raw member content.

Archive item 35: UDCT_v714_Protocol01_NextStage_DesignGate_20261029_20261110_124522_601.zip

- SHA-256: `79324efed04d9c23a0d51a8f4fd0ac4c9aece335abfb2cfe1162bfe8fd9bd69f`; bytes: 10553; nested CRC: pass

- Contained files: `UDCT_v714_Protocol01_NextStage/`, `UDCT_v714_Protocol01_NextStage/PROTOCOL01_NEXT_STAGE_DESIGN.md`, `UDCT_v714_Protocol01_NextStage/BASELINE_VERIFICATION_ARCHIVE.zip`

This ledger entry contains no directly quoted technical note in the prior reconstruction; consult the companion archive for raw member content.

Archive item 36: UDCT_v714_Protocol01_OM_NecessaryGate_20261029_20261110_124531_515.zip

- SHA-256: `6e3b7e43547ff358334000bc5c4e58d94a6c442543a7debfe09d5bc48ab0d51a`; bytes: 3796; nested CRC: pass

- Contained files: `threshold_scan.csv`, `om_necessary_gate.py`, `README.md`, `results.json`, `run_output.txt`

Source excerpt — README.md (excerpt):

UDCT v7.14 — OM necessary-condition gate (2026-10-29)

New calculation, exploratory only. Not preregistered confirmatory evidence.

For tracer density `rho_p(x)=(1+x^p)^(-5/p)` and OM augmented density

`rho_Q(x)=(1+x^2/q^2)rho_p(x)`, where `x=r/a`, `q=r_a/a`, a necessary condition for a nonnegative OM distribution function is that augmented density decrease outward (equivalently increase with relative potential, where that convention exis

Differentiation yields the necessary inequality

`q^2 >= (2/5)x^(2-p) - (3/5)x^2` for every `x>0`.

For `0<p<2`, maximizing the right-hand side gives

`x_* = ((2-p)/3)^(1/p)` and `q_min = sqrt((p/5)*x_*^(2-p))`.

At `p=1.5`, this gives the value in `results.json`. The old exploratory candidate `q≈0.4664667874` passes this *necessary* gate, which neither confirms it as a true threshold nor proves DF positivity.

This inequality depends only on the tracer and OM anisotropy parametrization, not on the detailed gravitational acceleration, provided the potential is monotonic. It does **not** resolve the isolated MOND infinite-potential convention. The

Run `python om_necessary_gate.py` (requires numpy, scipy). Output includes analytic-vs-optimization cross-checks. The CSV scans p; it is not an observational fit. No stability or Hydrus I star-level likelihood is claimed.

Source excerpt — results.json (excerpt):

{"p": 1.5, "x_at_max": 0.3028534321386899, "necessary_ra_over_a_min": 0.40632051688873816, "prior_exploratory_candidate": 0.4664667874, "candidate_passes_necessary_gate": true, "status": "necessary condition only; not OM DF inversion"}

Archive item 37: UDCT_v714_Protocol01_Prereg_Candidate_20261109_20261110_124944_011.zip

- SHA-256: `13c8136a13630672885293435e2c89b97d897e6d8bafb3a6445fc8cb14cf30ab`; bytes: 6162; nested CRC: pass

- Contained files: `UDCT_v714_Prereg_Protocol01_20261109/PROTOCOL.md`, `UDCT_v714_Prereg_Protocol01_20261109/README.md`, `UDCT_v714_Prereg_Protocol01_20261109/VALIDATION_LOG.txt`, `UDCT_v714_Prereg_Protocol01_20261109/protocol_config.json`, `UDCT_v714_Prereg_Protocol01_20261109/requirements.txt`, `UDCT_v714_Prereg_Protocol01_20261109/validate_protocol.py`

Source excerpt — UDCT_v714_Prereg_Protocol01_20261109/README.md (excerpt):

UDCT v7.14 Protocol 01 — preregistration candidate

This ZIP contains a **candidate protocol**, not an already published preregistration, and **no new prospective scientific results**. It is prepared to lock the next tests while preserving the earlier Steps 01–50 and Audits 01–06.

Run `python validate_protocol.py` to check protocol metadata and print SHA-256 hashes. This script deliberately does NOT evaluate the QUMOND/OM model. Review the unresolved physical potential/energy boundary gate in `PROTOCOL.md` before for

Historical values: p=1.5 local leading-term cancellation; ra/a~0.466466787396 local finite-window contact. Do not equate either with a global universal boundary. Direct individual Hydrus I stellar-velocity fitting has not been performed.

Source excerpt — UDCT_v714_Prereg_Protocol01_20261109/protocol_config.json (excerpt):

{"status": "PROPOSED LOCK; NOT PUBLISHED; NO PROSPECTIVE RUNS", "version": "UDCT v7.14 Protocol 01 preregistration candidate 2026-11-09", "historical_exploratory": {"p_local": 1.5, "ra_over_a_local": 0.46646678739601644, "energy_label_r_ove

Archive item 38: UDCT_v714_Protocol01_Preregistration_DRAFT_20261110_124428_397.md

- SHA-256: `aa2e6ed3b5cd8fd3ef9131740d5c28fb5cc1b1ff724e6a27a9575568e0169613`; bytes: 4336; nested CRC: N/A

- Contained files:

Source excerpt — UDCT_v714_Protocol01_Preregistration_DRAFT_20261110_124428_397.md (excerpt):

UDCT v7.14 Formal Preregistration — Protocol 01 (DRAFT)

Status: Draft only. Not preregistered, frozen, executed, or published. 29 October 2026 (NZ).

Provenance and separation

This protocol was written after inspection of archived exploratory Step 40, Step 50, Audit 03, and Audit 06. All values in the next section are **prior exploratory findings**, not prospective predictions. The three original archives remain

Archived exploratory results

- Step 40: isolated spherical simple-QUMOND with Plummer gravitating mass; G=4.30091e-3 pc (km/s)^2/Msun; M=32730 Msun; a=53.3 pc; a0=1.2e-10 m/s^2; tracer rho_p(r)=[1+(r/a)^p]^(-5/p).

- Step 40: central Psi(r)=A r^(3/2)+(k/4)r^2+...; at p=1.5, leading rho'' coefficient cancels and next-order isotropic f(E) has a positive E^(-1/6) term under stated assumptions.

- Step 50: OM local critical ra/a=0.46646678739601644 at p=1.5; energy-label r/a=0.47409755283222904; search interval r/a in [0.25,0.75]. This is not global DF positivity.

- Audit 03: local p sweep [1.49,1.495,1.499,1.5,1.501,1.505] with common Step47 kernel.

- Audit 06: bounded/golden optimizers shared the same physical kernel. Agreement is not independent physical validation.

Research questions

RQ1: Can an independently derived asymptotic expansion reproduce the sign and exponent of the p=1.5 subleading term, without copying the Step 40 derivation?

RQ2: Can an independently implemented OM inversion reproduce the **local** contact boundary and determine whether other negative lobes occur on a separately defined extended energy domain?

RQ3: How do independent inversion methods and quadrature convergence affect classifications of DF nonnegativity?

Primary and secondary outcomes

Primary: signed difference between independently derived and archived asymptotic coefficients and powers; local critical ra/a discrepancy between independently implemented inversion and the archived Step 50 benchmark; signs of DF with numer

Secondary: sensitivity to quadrature order, energy grid, optimization method, and tracer exponent.

Archive item 39: UDCT_v714_Protocol01_Reproducibility_20261110_124610_259.zip

- SHA-256: `c6f1c48175175947563d32b9b4e6dcf8480d89dd24e8f4960959514d043c5d4b`; bytes: 3038; nested CRC: pass

- Contained files: `UDCT_v714_Protocol01/`, `UDCT_v714_Protocol01/README.md`, `UDCT_v714_Protocol01/UDCT_v714_Protocol01_Asymptotic_Audit.py`, `UDCT_v714_Protocol01/UDCT_v714_Protocol01_Config.json`, `UDCT_v714_Protocol01/UDCT_v714_Protocol01_Results.csv`

Source excerpt — UDCT_v714_Protocol01/README.md (excerpt):

UDCT v7.14 — Protocol 01: Independent central asymptotic audit

This **pre-registration preparation** package checks the algebra of the *assumed* leading-order forms

- `psi(r) = A r^(3/2)`

- `rho(r) = rho0 - C r^p`

where `A>0`, `C>0`, `p>0`, and `psi>0` is a potential difference measured from the center.

It independently differentiates `rho(psi)` twice with SymPy, checks the result against a separately specified expression, and writes an exact rational-sign CSV at `A=C=psi=1`.

Run

Requires Python 3.9+ and `sympy`.

~~~bash

python -m pip install sympy

python UDCT_v714_Protocol01_Asymptotic_Audit.py

~~~

Expected identity:

Source excerpt — UDCT_v714_Protocol01/UDCT_v714_Protocol01_Config.json (excerpt):

{"protocol": "UDCT v7.14 Protocol 01: independent leading-order asymptotic algebra audit", "status": "pre-registration preparation; not a post-registration test", "assumptions": ["psi(r)=A*r^(3/2)", "rho(r)=rho0-C*r^p", "A>0, C>0, p>0, psi>

Archive item 40: UDCT_v714_Protocol01_Reproducibility_REISSUED_20261029_20261110_124622_899.zip

- SHA-256: `c6f1c48175175947563d32b9b4e6dcf8480d89dd24e8f4960959514d043c5d4b`; bytes: 3038; nested CRC: pass

- Contained files: `UDCT_v714_Protocol01/`, `UDCT_v714_Protocol01/README.md`, `UDCT_v714_Protocol01/UDCT_v714_Protocol01_Asymptotic_Audit.py`, `UDCT_v714_Protocol01/UDCT_v714_Protocol01_Config.json`, `UDCT_v714_Protocol01/UDCT_v714_Protocol01_Results.csv`

Source excerpt — UDCT_v714_Protocol01/README.md (excerpt):

UDCT v7.14 — Protocol 01: Independent central asymptotic audit

This **pre-registration preparation** package checks the algebra of the *assumed* leading-order forms

- `psi(r) = A r^(3/2)`

- `rho(r) = rho0 - C r^p`

where `A>0`, `C>0`, `p>0`, and `psi>0` is a potential difference measured from the center.

It independently differentiates `rho(psi)` twice with SymPy, checks the result against a separately specified expression, and writes an exact rational-sign CSV at `A=C=psi=1`.

Run

Requires Python 3.9+ and `sympy`.

~~~bash

python -m pip install sympy

python UDCT_v714_Protocol01_Asymptotic_Audit.py

~~~

Expected identity:

Source excerpt — UDCT_v714_Protocol01/UDCT_v714_Protocol01_Config.json (excerpt):

{"protocol": "UDCT v7.14 Protocol 01: independent leading-order asymptotic algebra audit", "status": "pre-registration preparation; not a post-registration test", "assumptions": ["psi(r)=A*r^(3/2)", "rho(r)=rho0-C*r^p", "A>0, C>0, p>0, psi>

Archive item 41: UDCT_v714_Protocol01_Verification_REISSUED_20261029_20261110_124508_583.zip

- SHA-256: `e929e3aa0577b1cbd9276ba5656c8cce8d4d87b32edcc1f6dcfdf8ca42f1a079`; bytes: 7961; nested CRC: pass

- Contained files: `UDCT_v714_Protocol01_Verification_Package/`, `UDCT_v714_Protocol01_Verification_Package/verify.py`, `UDCT_v714_Protocol01_Verification_Package/UDCT_v714_Protocol01_Independence_Audit.md`, `UDCT_v714_Protocol01_Verification_Package/README.md`, `UDCT_v714_Protocol01_Verification_Package/UDCT_v714_Protocol01_Verification_Results.csv`, `UDCT_v714_Protocol01_Verification_Package/UDCT_v714_Protocol01_Asymptotic_Notes.md`, `UDCT_v714_Protocol01_Verification_Package/UDCT_v714_Protocol01_OM_Test_Plan.md`, `UDCT_v714_Protocol01_Verification_Package/UDCT_v714_Protocol01_Independent_Series_Inversion.py`, `UDCT_v714_Protocol01_Verification_Package/UDCT_v714_Protocol01_Verification_Sum

Source excerpt — UDCT_v714_Protocol01_Verification_Package/README.md (excerpt):

UDCT v7.14 Protocol 01 — reconstructed verification package

Run `python -m pip install -r requirements.txt` and `python UDCT_v714_Protocol01_Independent_Series_Inversion.py`.

The script regenerates the CSV and JSON. The supplied Gemini message was incomplete and had a derivative bug; this is a corrected reconstruction, **not** an independently verified reproduction of all Step 40/47 results. No global OM inversi

Source excerpt — UDCT_v714_Protocol01_Verification_Package/UDCT_v714_Protocol01_Verification_Summary.json (excerpt):

{"status": "PASS: supplied toy-model algebra and sampled numerical checks", "model": "simple spherical MOND acceleration + Plummer gravitational mass + generalized tracer", "symbolic_coefficient": "4*B*C/(9*A**(7/3))", "expected_coefficient

Archive item 42: UDCT_v714_Sensitivity_Audit_Fresh_Link_20261029_20261110_124715_183.zip

- SHA-256: `44465f2e0c22492f5102d2621cf6d0a2fa29e876de7204ee0679c2518c38e82e`; bytes: 7694; nested CRC: pass

- Contained files: `UDCT_v714_Protocol01_Sensitivity_Audit_20261029/`, `UDCT_v714_Protocol01_Sensitivity_Audit_20261029/run_sensitivity.py`, `UDCT_v714_Protocol01_Sensitivity_Audit_20261029/README.md`, `UDCT_v714_Protocol01_Sensitivity_Audit_20261029/summary.json`, `UDCT_v714_Protocol01_Sensitivity_Audit_20261029/requirements.txt`, `UDCT_v714_Protocol01_Sensitivity_Audit_20261029/baseline_reference.py`, `UDCT_v714_Protocol01_Sensitivity_Audit_20261029/sensitivity.csv`, `UDCT_v714_Protocol01_Sensitivity_Audit_20261029/convergence_summary.csv`

Source excerpt — UDCT_v714_Protocol01_Sensitivity_Audit_20261029/README.md (excerpt):

UDCT v7.14 — Protocol 01, exploratory sensitivity audit (2026-10-29)

Status

This is a **pre-preregistration numerical audit**, not confirmatory evidence or a physically validated OM distribution function. It is NOT an observational fit to Hydrus I.

Run

Python 3.10+; `pip install -r requirements.txt`; `python run_sensitivity.py`.

Files

- `run_sensitivity.py`: reproduces 72 formal inversion-integral evaluations (4 anisotropy ratios × 3 radial/energy proxies × 3 outer cutoffs × 2 tolerances).

- `baseline_reference.py`: frozen earlier shared spherical simple-MOND field, density derivatives and interpolated central potential difference.

- `sensitivity.csv`: all 72 results.

- `convergence_summary.csv`: range and sign agreement in 12 groups.

- `summary.json`: machine-readable audit summary.

Fundamental limitation

The baseline potential is DeltaPhi(r)=Phi(r)-Phi(0), unbounded as r→infinity for isolated MOND. It is not automatically the finite OM relative potential Psi=Phi(infinity)-Phi(r). The computed quantity is therefore called `formal_integral`,

Interpretation

The 12/12 stable signs and maximum spread reflect the specified sampled numerical checks only. They do not cover the continuous energy interval, alternative boundary conditions, or independent model implementation. The local p=1.5 leading-o

Next design gate

Source excerpt — UDCT_v714_Protocol01_Sensitivity_Audit_20261029/summary.json (excerpt):

{"n_evaluations": 72, "n_groups": 12, "sign_stable_groups": 12, "max_abs_spread": 5.536572068864931e-12, "status": "EXPLORATORY NUMERICAL SENSITIVITY ONLY", "critical_limit": "Formal integral uses unbounded central potential difference; NOT

Archive item 43: UDCT_v714_source_inventory_20261110_124418_886.csv

- SHA-256: `9f8b166baf11ff178b59ed36d6e64fd2bb4a88f08b07acc32b409a477ce86d7a`; bytes: 18818; nested CRC: N/A

- Contained files:

Source excerpt — UDCT_v714_source_inventory_20261110_124418_886.csv (excerpt):

source,entry,bytes,nested_entries,nested_names,error

7.14 Exploratory audit 1~6.zip,UDCT_v7.14_Exploratory_Audit_01.zip,21861,22,UDCT_v7.14_Exploratory_Audit_01/ | UDCT_v7.14_Exploratory_Audit_01/original_step40/ | UDCT_v7.14_Exploratory_Audit_01/original_step40/audit.csv | UDCT_v7.14_Explora

7.14 Exploratory audit 1~6.zip,UDCT_v7.14_Exploratory_Audit_02.zip,14935,10,UDCT_v7.14_Exploratory_Audit_02/ | UDCT_v7.14_Exploratory_Audit_02/requirements.txt | UDCT_v7.14_Exploratory_Audit_02/__pycache__/ | UDCT_v7.14_Exploratory_Audit_02

7.14 Exploratory audit 1~6.zip,UDCT_v7.14_Exploratory_Audit_03.zip,6493,7,UDCT_v7.14_Exploratory_Audit_03/ | UDCT_v7.14_Exploratory_Audit_03/audit03.py | UDCT_v7.14_Exploratory_Audit_03/

Archive item 44: UDCT_v714_Step50_Kernel_Audit_20261109_20261110_124905_847.zip

- SHA-256: `7d923eff9c5ec71650974ac944bb9b8dcd24001701e9af20d989130ae93bea89`; bytes: 10698; nested CRC: pass

- Contained files: `README.md`, `audit_summary.json`, `requirements.txt`, `original_step50.py`, `original_summary.json`, `original_step47_reference.py`, `audit.py`, `derivative_crosscheck.csv`, `original_critical_tangencies.csv`

Source excerpt — README.md (excerpt):

UDCT v7.14 — Step 50 kernel audit (exploratory)

Run `pip install -r requirements.txt` then `python audit.py`. Outputs: `derivative_crosscheck.csv`, `audit_summary.json`.

This audits the **original Step 50 model** (Plummer gravitating source; tracer `[1+(r/a)^p]^(-5/p)`; OM radius in units of `a`). The original Step 50 source and its original outputs are included with `original_` prefixes for provenance.

Three independent checks at the **formula level**: (1) density derivatives from central finite differences in log-radius versus analytic expressions; (2) exact affine OM dependence on `1/(r_a/a)^2`; (3) enclosed Plummer mass by independent

Important: these tests do NOT independently reconstruct the physical QUMOND field, do NOT re-run the costly Step 50 DF inversion, and do NOT validate the global DF or observational Hydrus I velocities. The original 0.46646678739601644 i

Source excerpt — audit_summary.json (excerpt):

{"status": "exploratory code-level verification, NOT global OM positivity", "derivative_cases": 64, "max_derivative_relative_error": 5.997208065599098e-07, "max_affine_relative_error": 9.55036526671912e-15, "max_enclosed_mass_relative_error

Archive item 45: UDCT_v714_Step50_Source_Recovery_20261109_20261110_124858_579.zip

- SHA-256: `f169a3632e97eb4ec52d3e2b4bb5fea8f57337be7d33bd75f9fe6a0592642c5b`; bytes: 9922; nested CRC: pass

- Contained files: `p1p5_lobe_collapse.csv`, `independent_tracer_necessary_gate.py`, `source_manifest.json`, `step47_reference.py`, `execution_log.txt`, `critical_tangencies.csv`, `step50.py`, `necessary_gate_original_tracer.csv`, `README.md`, `summary.json`, `requirements.txt`

Source excerpt — README.md (excerpt):

UDCT v7.14 Step 50 original-source recovery and tracer check (2026-11-09)

Status

Source recovery and a **new analytic necessary-condition calculation**, not preregistered confirmatory research. Original Step 50 `summary.json` is preserved unchanged; its results are archived results, not rerun in this package.

Recovered model

The original Step 50 summary identifies a spherical isolated QUMOND/Plummer gravitational source, tracer density `rho(x)=[1+x^p]^(-5/p)`, Osipkov–Merritt (OM) anisotropy, and a local `r/a=0.25..0.75` tangency search. The original `step47_re

For p=1.5, archived Step 50 reports `r_a/a=0.46646678739601644`, tangency energy-label `r/a=0.47409755283222904`, and 96/192/384 Gauss-Legendre quadrature comparisons. This is a **local finite-window tangency**, not a global DF positivity b

Independent necessary condition for the ACTUAL tracer

For `x=r/a`, `rho=(1+x^p)^(-5/p)`, and `rho_aug=(1+x^2/s^2)*rho` with `s=r_a/a`, nonnegative OM DF in a monotonic relative potential requires `d rho_aug/dr<=0` (a necessary, not sufficient, condition). Algebra gives

`s^2 >= max_{x>0} [(2/5)*x^(2-p) - (3/5)*x^2]`.

For `0<p<2`, the maximizing `x=((2-p)/3)^(1/p)`. At `p=2`, the maximum occurs as `x->0`, giving `s>=sqrt(2/5)=0.6324555`. At `p=1.5`, this yields approximately `s>=0.4063205`. Thus **the original 0.4664668 candidate passes this particular n

Important cautions

- The earlier p=2 necessary-condition threshold 0.6324555 cannot be used to reject a p=1.5 tracer calculation.

- The isolated deep-MOND potential is not conventionally normalized to zero at infinity; the `outer=1e9` cutoff and DF inversion energy convention require a dedicated physical boundary audit.

- Source `step47_

Source excerpt — source_manifest.json (excerpt):

{"sources": [{"archive": "7.14 Hessian 26~50.zip", "members": 25, "sha256": "14da6463e2dc1b9af6726c448517aa6013709ae2e967a3b5cb8d7ae985292f58", "zip_integrity": true}, {"archive": "7.14 Hessian 1~25.zip", "members": 80, "sha256": "24e646cb9

Archive item 46: UDCT_v714_Tail_Adaptive_Crosscheck_REISSUED_20261030_20261110_124821_346.zip

- SHA-256: `5266e30d936033360808ad77fc4cf90d1300a94815ce17edc95809636e1f5532`; bytes: 4677; nested CRC: pass

- Contained files: `run_tail.py`, `step47_reference.py`, `tail_results.csv`, `tail_summary.json`, `README.md`

Source excerpt — README.md (excerpt):

UDCT v7.14 — Tail and adaptive quadrature exploratory audit

Date: 2026-10-30. Exploratory numerical follow-up; not preregistered confirmation.

Run `python run_tail.py` with numpy, scipy, pandas. `step47_reference.py` is the unmodified archived Step 47 kernel. The new `run_tail.py` compares 192/384-point Gaussian quadrature with adaptive scipy quad over the logarithmic radius, comp

All seven selected sample values were positive or near zero. This is **not** a proof of nonnegativity on a continuous interval or over all binding energies, and the adaptive implementation shares the same gravity and density derivative func

Next: independent symbolic derivatives, global minimization with certified numerical error, cutoff convergence and energy-domain endpoint asymptotics, then observational comparison as a separate phase.

Source excerpt — tail_summary.json (excerpt):

{"points": 7, "minimum_384": 5.572739167674104e-15, "negative_below_minus_1e_10": 0, "method": "adaptive quad cross-check shares physical gravity and rho2phi kernels; finite outer cutoff"}

Appendix B. Dialogue image review scope and representative contact sheet

The contact sheet below samples 11 distinct positions in the 1,642-image archive. It is provided for traceability, not as evidence of a complete visual review. The source JPEGs remain in the uploaded ZIP and should be reviewed at native resolution before any definitive chronology is published.

Appendix C. Reproducibility, interpretation and disclosure checklist

Open audit gap: 1,631 screenshots were not individually interpreted in this drafting pass. No claim is made that every dialogue assertion has been reconciled with a code output. This document is therefore an archival research DRAFT, not a completed final paper.

• Original ZIPs and ledger hashes preserved without alteration.

• Charts are previously generated from archived CSV outputs; they are not newly rerun solutions.

- No claim of full image-dialogue transcription or verified experiment chronology.
- No certified global f(Q) positivity or Q=0 endpoint conclusion.
- No direct individual-star likelihood fit to Hydrus I in this retrospective report.
- No Bullet Cluster merger or lensing simulation executed here.
- No proof of historical first discovery without systematic prior-art review.
- No newly preregistered result is inferred from retrospective exploratory data.